MNIST Object Detection Model Development and Improvement Using Roboflow and YOLO

1. Introduction

Object detection is a computer vision task that aims to identify both **what objects are present** in an image and **where they are located**. Unlike image classification, which assigns a single label to an entire image, object detection predicts one or more bounding boxes together with class labels for each detected object.

This project focuses on handwritten digit detection using a dataset derived from MNIST. The original MNIST dataset is one of the most widely used benchmark datasets in machine learning and is typically used for **image classification**, where a single handwritten digit appears in each image. However, in this project, the task was reformulated as an **object detection problem**, meaning that the model had to locate and classify multiple digit objects inside an image.

To accomplish this, Roboflow was used for dataset versioning, preprocessing, and augmentation management, while YOLO was used as the object detection model for training and evaluation. The purpose of this notebook is not only to train a detector, but also to document the full improvement process: starting from a baseline model, identifying a limitation in the default training setup, refining the augmentation strategy, and comparing multiple dataset versions to obtain a better final model.

2. Conceptual Background

2.1 Image Classification vs. Object Detection

Image classification and object detection solve related but fundamentally different problems.

In **image classification**, the model predicts one label for the entire image. For example, if an image contains a handwritten digit, the model predicts whether it is 0, 1, 2, or another class. In this setting, the model is not required to determine the location of the object.

In **object detection**, the model must predict:

- the class of each object
- the location of each object through a bounding box

Therefore, object detection is more complex than classification because it combines both recognition and localization.

In the context of this project, this distinction is important because MNIST is originally a classification dataset, while YOLO is an object detection model. As a result, the dataset had to be reorganized and exported in a detection-compatible format.

2.2 Why YOLO Was Used

YOLO, which stands for **You Only Look Once**, is a real-time object detection framework designed to perform object localization and classification in a single model pipeline. YOLO is widely used because it is efficient, relatively easy to train, and provides strong performance on detection tasks.

In this project, YOLO was chosen for three reasons:

1. It supports custom object detection datasets in a straightforward format.
2. It provides convenient training, validation, and prediction workflows.
3. It makes it easy to compare multiple experiments using the same model architecture and different dataset versions.

2.3 Why Roboflow Was Used

Roboflow was used as the dataset management platform for this project. Its role was especially important because the experiment involved multiple dataset versions rather than a single fixed dataset.

Roboflow was used for:

- annotation handling
- preprocessing
- augmentation setup
- dataset version generation
- export of training-ready object detection datasets

This version-based workflow made it possible to compare how different augmentation policies influenced final model performance.

3. Objective of the Study

The main objective of this study is to develop and improve a handwritten digit object detection model using Roboflow and YOLO.

More specifically, the study aims to:

1. establish a baseline detection model on the first dataset version
2. analyze the limitations of the default YOLO training configuration
3. test whether removing unsuitable augmentation improves digit detection performance

4. compare multiple dataset versions generated in Roboflow
5. identify the version that produces the most balanced object detection performance

Through this process, the notebook demonstrates not only final performance, but also the reasoning behind each improvement step.

4. Experimental Environment

Before conducting the experiments, the software environment was verified. The versions of Python, Ultralytics, and Roboflow were checked to ensure compatibility and reproducibility of the training pipeline.

The final training experiments were performed with GPU acceleration after confirming CUDA availability. This was important because multiple object detection experiments were conducted across several dataset versions, and GPU training significantly reduced execution time.

In [4]:

```
import ultralytics
import roboflow
import sys

print("Python:", sys.version)
print("Ultralytics:", ultralytics.__version__)
print("Roboflow:", roboflow.__version__)
```

```
Python: 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD
64)]
Ultralytics: 8.4.22
Roboflow: 1.2.16
```

5. Dataset Preparation and Structure Verification

Three dataset versions were prepared in advance:

- **V1**: initial baseline dataset
- **V2**: improved dataset with additional augmentation settings
- **V3**: further refined dataset version

Before training, the local directory structure and image counts for each dataset split were checked. This step was necessary to verify that all versions contained valid `train`, `valid`, and `test` folders and that the datasets were correctly exported from Roboflow.

In [5]:

```
from pathlib import Path

DATASETS = {
    "v1": Path("./data/mnist/mnist_v1"),
    "v2": Path("./data/mnist/mnist_v2"),
    "v3": Path("./data/mnist/mnist_v3"),
```

```python
}

IMAGE_EXTS = {".jpg", ".jpeg", ".png", ".bmp", ".webp"}

for name, path in DATASETS.items():
    print(f"\n===== {name} =====")
    print("resolved path:", path.resolve())
    print("exists:", path.exists())

    yaml_path = path / "data.yaml"
    train_img = path / "train" / "images"
    valid_img = path / "valid" / "images"
    test_img  = path / "test" / "images"

    print("data.yaml:", yaml_path.exists())
    print("train/images:", train_img.exists())
    print("valid/images:", valid_img.exists())
    print("test/images :", test_img.exists())

    def count_images(folder: Path):
        if not folder.exists():
            return 0
        return sum(1 for f in folder.iterdir() if f.suffix.lower() in IMAGE_EXTS)

    print("train image count:", count_images(train_img))
    print("valid image count:", count_images(valid_img))
    print("test  image count:", count_images(test_img))
```

```
===== v1 =====
resolved path: D:\workplace_00\data\mnist\mnist_v1
exists: True
data.yaml: True
train/images: True
valid/images: True
test/images : True
train image count: 35
valid image count: 10
test  image count: 5

===== v2 =====
resolved path: D:\workplace_00\data\mnist\mnist_v2
exists: True
data.yaml: True
train/images: True
valid/images: True
test/images : True
train image count: 104
valid image count: 10
test  image count: 5

===== v3 =====
resolved path: D:\workplace_00\data\mnist\mnist_v3
exists: True
data.yaml: True
train/images: True
valid/images: True
test/images : True
train image count: 105
valid image count: 10
test  image count: 5
```

In [ ]:
```python
yaml_path = Path("./data/mnist/mnist_v1/data.yaml")
print(yaml_path.read_text(encoding="utf-8"))
```


```
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 10
names: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']

roboflow:
  workspace: jungkyus-workspace
  project: my-first-project-kh8et
  version: 1
  license: Public Domain
  url: https://universe.roboflow.com/jungkyus-workspace/my-first-project-kh8et/datas
et/1
```


Local YAML Configuration for Training

To ensure stable training in the local environment, a new `data_local.yaml` file was generated for each dataset version. In this configuration, the dataset paths were rewritten as follows:

- `train: train/images`
- `val: valid/images`
- `test: test/images`

This modification prevented path resolution errors during YOLO training and made the notebook portable across local executions.

In [7]:
```
from pathlib import Path
import yaml

DATASET_ROOTS = {
    "v1": Path("./data/mnist/mnist_v1"),
    "v2": Path("./data/mnist/mnist_v2"),
    "v3": Path("./data/mnist/mnist_v3"),
}

for version_name, root in DATASET_ROOTS.items():
    orig_yaml = root / "data.yaml"
    local_yaml = root / "data_local.yaml"

    if not orig_yaml.exists():
        print(f"[{version_name}] data.yaml not found: {orig_yaml.resolve()}")
        continue

    with open(orig_yaml, "r", encoding="utf-8") as f:
        data = yaml.safe_load(f)

    data["train"] = "train/images"
    data["val"] = "valid/images"
    data["test"] = "test/images"

    with open(local_yaml, "w", encoding="utf-8") as f:
        yaml.safe_dump(data, f, allow_unicode=True, sort_keys=False)

    print(f"\n[{version_name}] saved -> {local_yaml.resolve()}")
    print(local_yaml.read_text(encoding="utf-8"))
```

```
[v1] saved -> D:\workplace_00\data\mnist\mnist_v1\data_local.yaml
```

```
train: train/images
val: valid/images
test: test/images
nc: 10
names:
- '0'
- '1'
- '2'
- '3'
- '4'
- '5'
- '6'
- '7'
- '8'
- '9'
roboflow:
  workspace: jungkyus-workspace
  project: my-first-project-kh8et
  version: 1
  license: Public Domain
  url: https://universe.roboflow.com/jungkyus-workspace/my-first-project-kh8et/datas
et/1
```


```
[v2] saved -> D:\workplace_00\data\mnist\mnist_v2\data_local.yaml
```

```
train: train/images
val: valid/images
test: test/images
nc: 10
names:
- '0'
- '1'
- '2'
- '3'
- '4'
- '5'
- '6'
- '7'
- '8'
- '9'
roboflow:
  workspace: jungkyus-workspace
  project: my-first-project-kh8et
  version: 2
  license: Public Domain
  url: https://universe.roboflow.com/jungkyus-workspace/my-first-project-kh8et/datas
et/2
```


```
[v3] saved -> D:\workplace_00\data\mnist\mnist_v3\data_local.yaml
```

```
train: train/images
val: valid/images
test: test/images
nc: 10
names:
```

```
- '0'
- '1'
- '2'
- '3'
- '4'
- '5'
- '6'
- '7'
- '8'
- '9'
roboflow:
  workspace: jungkyus-workspace
  project: my-first-project-kh8et
  version: 3
  license: Public Domain
  url: https://universe.roboflow.com/jungkyus-workspace/my-first-project-kh8et/datas
et/3
```


6. Dataset Versions Used in This Study

Three dataset versions were prepared and used throughout the experiments. Each version reflects a different stage of the dataset design process.

Version 1 (V1): Initial Baseline Dataset

Version 1 represents the original baseline dataset exported from Roboflow. This version was used to establish the initial reference performance of the model.

Preprocessing

- Auto-Orient applied

Augmentation

- No augmentation applied

V1 was intentionally kept simple in order to provide a clean baseline. It was first trained with the default YOLO configuration and later reused in the no-flip experiment to isolate the effect of the training policy.

Version 2 (V2): Augmented Intermediate Dataset

Version 2 was created by adding augmentation settings in Roboflow. Compared with V1, this version increased the number of training images and introduced more variation into the training data.

Preprocessing

- Auto-Orient applied

- Resize: Stretch to 640×640

Augmentation

- Outputs per training example: 3
- Rotation: between -15° and +15°

Compared with V1, Version 2 increased the number of training images and introduced geometric variation through rotation. The purpose of this version was to improve robustness to orientation changes while preserving the overall semantic structure of handwritten digits.

Version 3 (V3): Final Refined Dataset

Version 3 was the most refined dataset version used in this study. It built upon Version 2 by adding appearance-based augmentations in addition to rotation.

Preprocessing

- Auto-Orient applied
- Resize: Stretch to 640×640

Augmentation

- Outputs per training example: 3
- Rotation: between -15° and +15°
- Brightness: between -15% and +15%
- Exposure: between -15% and +15%
- Blur: up to 2.5 px

The purpose of Version 3 was to improve robustness not only to geometric variation but also to visual quality variation. By introducing brightness, exposure, and blur adjustments, this version aimed to make the detector more stable under less ideal visual conditions.

Rationale for Version Progression

The three versions reflect a progressive experimental strategy:

- **V1** established the baseline using minimal preprocessing and no augmentation.
- **V2** introduced moderate geometric augmentation through resizing and rotation.
- **V3** extended this strategy by adding visual-condition augmentations such as brightness, exposure, and blur.

This progressive version design made it possible to analyze how controlled dataset refinement influenced final detection performance.

An important aspect of this study is that the dataset was not treated as fixed. Instead, multiple Roboflow versions were created with progressively refined preprocessing and augmentation settings. This allowed the experiment to evaluate not only model behavior, but also the effect of controlled dataset design choices.

7. Evaluation Metrics

To evaluate the object detection models, four main metrics were used throughout the experiments:

- **Precision**
- **Recall**
- **mAP50**
- **mAP50-95**

These metrics are standard in object detection, but they reflect different aspects of model performance. Therefore, understanding their meaning is essential for interpreting the experimental results.

7.1 Precision

Precision measures how many predicted detections are actually correct.

In simple terms, precision answers the question:

> When the model says there is a digit, how often is it correct?

A higher precision means that the model produces fewer false positives. In this project, high precision is important because incorrect digit predictions can reduce the reliability of the detector, especially when multiple digits appear in the same image.

7.2 Recall

Recall measures how many real objects in the image are successfully detected by the model.

In simple terms, recall answers the question:

> Out of all real digits present in the image, how many did the model find?

A higher recall means that the model misses fewer objects. In this project, recall is important because the detector should identify as many digit instances as possible, especially when several digits are present in one image.

7.3 mAP50

mAP50 stands for mean Average Precision at an Intersection over Union threshold of 0.50.

This metric evaluates whether the predicted bounding box overlaps sufficiently with the ground-truth bounding box at a relatively lenient threshold. A higher mAP50 indicates that the model is performing well in terms of both classification and rough localization.

In this experiment, mAP50 is useful for understanding the model's general detection capability.

7.4 mAP50-95

mAP50-95 is a stricter and more comprehensive metric. It averages performance over multiple IoU thresholds from 0.50 to 0.95.

This means the metric evaluates not only whether the model detects the object, but also how accurately it localizes the object. Because of its stricter nature, mAP50-95 is usually lower than mAP50.

In this project, mAP50-95 is especially important because it reflects the overall quality of the final detector more reliably than a single-threshold metric.

7.5 Meaning of These Metrics in This Study

In the context of this project, each metric has a specific interpretive role:

- **Precision** indicates how reliable the detections are
- **Recall** indicates how many digit objects are successfully found
- **mAP50** indicates the overall detection quality at a standard threshold
- **mAP50-95** indicates the overall robustness and localization quality of the detector

Because this project compares several experimental conditions, the metrics should be interpreted together rather than individually. For example:

- a model with higher precision but lower recall may be more conservative
- a model with higher recall may detect more objects but also introduce more errors
- a model with higher mAP50-95 is generally considered more accurate and balanced overall

For this reason, the final model selection in this notebook is based on the combined interpretation of all four metrics rather than a single score alone.

8. Experimental Design Logic

The experiments in this notebook were designed as a step-by-step improvement process rather than a set of unrelated training trials.

The progression was as follows:

1. **V1 baseline**: establish the initial reference performance
2. **V1 no-flip**: test whether removing an unsuitable augmentation improves results
3. **V2 no-flip**: evaluate the effect of a more diverse augmented dataset
4. **V3 no-flip**: evaluate the final refined dataset version

This design makes it possible to separate two types of improvement:

- improvement caused by **training policy adjustment**
- improvement caused by **dataset version refinement**

As a result, the notebook captures not only which model performed best, but also why the performance improved over time.

9. Experiment 1: V1 Baseline Training

The first experiment used **Version 1** of the dataset as the baseline condition. This experiment was designed to establish the initial performance level of YOLO before introducing any task-specific correction.

At this stage, the model was trained using the default YOLO training behavior. This means the experiment reflects how a general-purpose object detection pipeline performs on the MNIST digit detection task without additional adjustment for domain-specific characteristics.

This baseline is important because all later experiments are evaluated relative to this starting point.

In [ ]:

```python
from ultralytics import YOLO

model_v1 = YOLO("yolov8m.pt")

results_v1 = model_v1.train(
    data="./data/mnist/mnist_v1/data_local.yaml",
    epochs=100,
    patience=50,
    imgsz=640,
    batch=8,
    project="./runs/mnist",
    name="v1_baseline",
    exist_ok=True
)

best_model = YOLO("./runs/detect/runs/mnist/v1_baseline/weights/best.pt")

metrics = best_model.val(
    data="./data/mnist/mnist_v1/data_local.yaml",
    split="val",
    device=0
)

precision = metrics.box.mp
```

```
recall = metrics.box.mr
map50 = metrics.box.map50
map5095 = metrics.box.map

print("\n===== V1 No-Flip Validation Metrics =====")
print(f"Precision   : {precision:.4f}")
print(f"Recall      : {recall:.4f}")
print(f"mAP50       : {map50:.4f}")
print(f"mAP50-95    : {map5095:.4f}")
```

```
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment
=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_
mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=./data/mnist/mnist_v1/data_local.yaml, degrees=0.0, d
eterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=N
one, end2end=None, epochs=100, erasing=0.4, exist_ok=True, fliplr=0.5, flipud=0.0, f
ormat=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hs
v_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0
=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=v1_baseline, nbs=64, nms=False, op
set=None, optimize=False, optimizer=auto, overlap_mask=True, patience=50, perspectiv
e=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=./runs/mnist,
rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, s
ave_crop=False, save_dir=D:\workplace_00\runs\detect\runs\mnist\v1_baseline, save_fr
ames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shea
r=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True,
single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=Non
e, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=
0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module                                       a
rguments
  0                  -1  1      1392  ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]
  1                  -1  1     41664  ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]
  2                  -1  2    111360  ultralytics.nn.modules.block.C2f
[96, 96, 2, True]
  3                  -1  1    166272  ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]
  4                  -1  4    813312  ultralytics.nn.modules.block.C2f
[192, 192, 4, True]
  5                  -1  1    664320  ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]
  6                  -1  4   3248640  ultralytics.nn.modules.block.C2f
[384, 384, 4, True]
  7                  -1  1   1991808  ultralytics.nn.modules.conv.Conv
[384, 576, 3, 2]
  8                  -1  2   3985920  ultralytics.nn.modules.block.C2f
[576, 576, 2, True]
  9                  -1  1    831168  ultralytics.nn.modules.block.SPPF
[576, 576, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  2   1993728  ultralytics.nn.modules.block.C2f
[960, 384, 2]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
```

```
 15                  -1  2    517632  ultralytics.nn.modules.block.C2f
[576, 192, 2]
 16                  -1  1    332160  ultralytics.nn.modules.conv.Conv
[192, 192, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  2   1846272  ultralytics.nn.modules.block.C2f
[576, 384, 2]
 19                  -1  1   1327872  ultralytics.nn.modules.conv.Conv
[384, 384, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  2   4207104  ultralytics.nn.modules.block.C2f
[960, 576, 2]
 22        [15, 18, 21]  1   3781486  ultralytics.nn.modules.head.Detect
[10, 16, None, [192, 384, 576]]
Model summary: 170 layers, 25,862,110 parameters, 25,862,094 gradients, 79.1 GFLOPs

Transferred 469/475 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.00.0 ms, read: 23.05.6 MB/s, size: 4.6 KB)
train: Scanning D:\workplace_00\data\mnist\mnist_v1\train\labels.cache... 35 images,
0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 35/35  0.0s
val: Fast image access  (ping: 0.10.0 ms, read: 21.88.9 MB/s, size: 5.3 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
rmining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000714, momentum=0.9) with parameter groups 77 weight(decay=0.
0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Plotting labels to D:\workplace_00\runs\detect\runs\mnist\v1_baseline\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to D:\workplace_00\runs\detect\runs\mnist\v1_baseline
Starting training for 100 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      3.64G      2.582      4.583      1.904         14        640: 100%
━━━━━━━━━━━━ 5/5 1.7it/s 3.0s0.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 3.4it/s 0.3s
                   all         10         54      0.011      0.182       0.02    0.0
0447

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100      3.65G       2.61      4.994      1.878         10        640: 100%
━━━━━━━━━━━━ 5/5 5.1it/s 1.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54     0.0113      0.296     0.0235    0.0
0561

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100       3.8G      2.247      4.112      1.659         26        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.2it/s 0.1s
                   all         10         54      0.023      0.255      0.037     0.
0108

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100      3.82G      1.737      3.275      1.244         24        640: 100%
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.7it/s 0.1s
                   all         10         54      0.033      0.485     0.0566     0.
0175

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      3.82G      1.714      3.067      1.253         15        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54     0.0594      0.126     0.0558     0.
0204

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100      3.82G      1.574      2.804      1.207         14        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.0it/s 0.1s
                   all         10         54      0.156      0.113     0.0492
0.016

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100      3.82G      1.541      2.537      1.209         22        640: 100%
━━━━━━━━━━━━ 5/5 5.2it/s 1.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.2it/s 0.1s
                   all         10         54      0.135      0.245      0.141     0.
0513

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100      3.82G      1.548      2.486      1.198         25        640: 100%
━━━━━━━━━━━━ 5/5 4.9it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.5it/s 0.1s
                   all         10         54      0.113       0.44      0.208     0.
0864

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100      3.82G      1.521      2.599      1.183         18        640: 100%
━━━━━━━━━━━━ 5/5 4.5it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.345      0.387      0.258
0.12

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      3.82G      1.589      2.404      1.245         26        640: 100%
```

```
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.234      0.419       0.33
0.158

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100      3.82G      1.493        2.2      1.199         37        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.365      0.527      0.456
0.235

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     12/100      3.82G      1.465      2.641       1.18         15        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.803      0.287       0.41
0.189

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     13/100      3.82G      1.561       2.44      1.237         32        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.492      0.408      0.456
0.253

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     14/100      3.82G      1.467      2.241       1.17         25        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.492      0.408      0.456
0.253

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     15/100      3.82G      1.585      2.433      1.298          7        640: 100%
━━━━━━━━━━━━ 5/5 5.0it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.374      0.621      0.464
0.265

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     16/100      3.82G      1.607       2.11      1.249         34        640: 100%
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.353      0.619      0.524
0.266

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     17/100      3.82G       1.47      2.044      1.217         18        640: 100%
```

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━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.6it/s 0.1s
                   all         10         54      0.462       0.63      0.622
0.361

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     18/100      3.82G      1.522       2.17      1.257         11        640: 100%
━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.462       0.63      0.622
0.361

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     19/100      3.82G      1.372      2.097      1.222         16        640: 100%
━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.559      0.431      0.459
0.249

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     20/100      3.82G      1.394      2.222      1.222         22        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.555      0.423      0.538
0.307

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     21/100      3.82G      1.471      2.035      1.207         25        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.555      0.423      0.538
0.307

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     22/100      3.82G      1.369      1.974      1.182         14        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.561      0.549      0.598
0.325

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     23/100      3.82G      1.487      2.022      1.217         28        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.457      0.691      0.678
0.356

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     24/100      3.82G      1.419      1.802      1.177         30        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.457      0.691      0.678
0.356

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     25/100      3.82G      1.481       1.83      1.206         28        640: 100%
━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.9it/s 0.1s
                   all         10         54       0.58      0.639       0.71
0.424

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     26/100      3.82G       1.49      1.861      1.224         33        640: 100%
━━━━━━━━━━━━ 5/5 4.5it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54       0.58      0.639       0.71
0.424

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     27/100      3.82G      1.443      1.675      1.185         29        640: 100%
━━━━━━━━━━━━ 5/5 4.7it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.421      0.735      0.694
0.366

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     28/100      3.82G      1.365      1.603      1.138         29        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.635      0.788      0.741
0.438

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     29/100      3.82G      1.469      1.736      1.283         17        640: 100%
━━━━━━━━━━━━ 5/5 4.9it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.635      0.788      0.741
0.438

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     30/100      3.82G      1.573      1.633      1.299         18        640: 100%
━━━━━━━━━━━━ 5/5 4.9it/s 1.0s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54        0.5      0.738      0.719
0.44

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     31/100      3.82G      1.422      1.605      1.241         23        640: 100%
```

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━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.507      0.724      0.723
0.417

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     32/100      3.82G      1.438      1.683      1.248         21        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.507      0.724      0.723
0.417

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     33/100      3.82G      1.408      1.648      1.203         28        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.464      0.726      0.658
0.353

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     34/100      3.82G      1.438      1.556      1.227         15        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.464      0.726      0.658
0.353

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     35/100      3.82G       1.36      1.434      1.166         22        640: 100%
━━━━━━━━━━━━ 5/5 4.5it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.579      0.685      0.675
0.368

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     36/100      3.82G       1.44      1.593      1.229         32        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.558      0.863      0.806
0.474

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     37/100      3.82G      1.353      1.468      1.214         28        640: 100%
━━━━━━━━━━━━ 5/5 4.9it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.558      0.863      0.806
0.474

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     38/100      3.82G      1.359      1.394      1.194         17        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.745      0.813      0.818
0.492

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     39/100      3.82G      1.382      1.513      1.242         29        640: 100%
━━━━━━━━━━━━ 5/5 4.7it/s 1.1s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.708      0.915       0.83
0.495

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     40/100      3.82G        1.4      1.403      1.217         29        640: 100%
━━━━━━━━━━━━ 5/5 4.4it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.708      0.915       0.83
0.495

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     41/100      3.82G       1.42      1.422      1.306          9        640: 100%
━━━━━━━━━━━━ 5/5 4.3it/s 1.2s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.602       0.91      0.819
0.462

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     42/100      3.82G      1.357      1.389      1.197         25        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.602       0.91      0.819
0.462

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     43/100      3.82G      1.371      1.425      1.133         35        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.2it/s 0.1s
                   all         10         54      0.586      0.905      0.866
0.482

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     44/100      3.82G      1.307      1.264      1.154         14        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.736      0.778      0.773
0.39

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     45/100      3.82G      1.417      1.304      1.193         37        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.736      0.778      0.773
0.39

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     46/100      3.82G      1.432      1.345      1.268         19        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.778      0.811      0.817
0.46

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     47/100      3.82G      1.376      1.341      1.212         13        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.802      0.762      0.826
0.487

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     48/100      3.82G      1.381      1.293       1.21         23        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.802      0.762      0.826
0.487

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     49/100      3.82G      1.406      1.295      1.245         10        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54      0.684      0.884      0.864
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     50/100      3.82G      1.319      1.296      1.175         26        640: 100%
━━━━━━━━━━━━ 5/5 4.5it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.684      0.884      0.864
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     51/100      3.82G      1.292      1.265      1.206         28        640: 100%
━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.1it/s 0.1s
                   all         10         54      0.785       0.88      0.944
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100      3.82G      1.326      1.155      1.179         31        640: 100%
```

```
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.761      0.905      0.905
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100      3.82G      1.303      1.223      1.208         18        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.9it/s 0.1s
                   all         10         54      0.761      0.905      0.905
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100      3.82G      1.419      1.387      1.225         10        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.778      0.893      0.893
0.544

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100      3.82G      1.363      1.192      1.204         17        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.861      0.803      0.951
0.594

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100      3.82G      1.323      1.186      1.151         25        640: 100%
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.861      0.803      0.951
0.594

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     57/100      3.82G      1.264      1.097      1.167         38        640: 100%
━━━━━━━━━━━━ 5/5 4.9it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.741      0.886      0.881
0.542

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     58/100      3.82G      1.267      1.093      1.158         13        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.741      0.886      0.881
0.542

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     59/100      3.82G      1.339      1.186      1.176         21        640: 100%
```

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━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.752      0.939      0.889
0.53

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     60/100      3.82G      1.402      1.143      1.185         29        640: 100%
━━━━━━━━━━━━ 5/5 4.5it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.771      0.936      0.964
0.599

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     61/100      3.82G      1.326      1.072      1.201         19        640: 100%
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.771      0.936      0.964
0.599

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     62/100      3.82G      1.327      1.145      1.213         22        640: 100%
━━━━━━━━━━━━ 5/5 4.8it/s 1.0s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.909      0.796      0.881
0.553

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     63/100      3.82G      1.306       1.13       1.18         12        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54       0.93      0.748      0.882
0.535

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     64/100      3.82G      1.393      1.155      1.226         19        640: 100%
━━━━━━━━━━━━ 5/5 4.6it/s 1.1s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.9it/s 0.1s
                   all         10         54       0.93      0.748      0.882
0.535

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     65/100      3.82G       1.31      1.126      1.191         14        640: 100%
━━━━━━━━━━━━ 5/5 5.1it/s 1.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.867      0.895      0.921
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     66/100      3.82G      1.213      1.109      1.123         28        640: 100%
```

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━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.867      0.895      0.921
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     67/100      3.82G      1.259      1.086      1.186         28        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.862      0.858      0.918
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     68/100      3.82G      1.237      1.016       1.16         24        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.858      0.851      0.911
0.547

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     69/100      3.82G       1.33      1.053      1.146         42        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.858      0.851      0.911
0.547

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     70/100      3.82G      1.248     0.9907      1.116         35        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.871      0.876      0.926
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     71/100      3.82G      1.243      1.073      1.174         33        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.7it/s 0.1s
                   all         10         54      0.853      0.894       0.92
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     72/100      3.82G       1.29      1.092       1.23         24        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.853      0.894       0.92
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     73/100      3.82G      1.227      1.036      1.106         21        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.3it/s 1.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.3it/s 0.1s
                   all         10         54      0.847      0.919      0.915
0.544

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     74/100      3.82G       1.28      1.009      1.158         23        640: 100%
━━━━━━━━━━━━ 5/5 4.7it/s 1.1s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.847      0.919      0.915
0.544

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     75/100      3.82G      1.222      0.979      1.195         12        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.881      0.895      0.912
0.513

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     76/100      3.82G      1.208      1.009      1.138         18        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.798      0.911      0.952
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     77/100      3.82G      1.283      0.966      1.177         26        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.798      0.911      0.952
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     78/100      3.82G      1.226     0.9698      1.127         18        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.849      0.842      0.951
0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     79/100      3.82G      1.184     0.8973        1.1         19        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.7it/s 0.1s
                   all         10         54      0.895      0.824      0.943
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     80/100      3.82G      1.204     0.9688      1.129         18        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.895      0.824      0.943
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     81/100      3.82G      1.222     0.9231      1.117         32        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.903       0.83      0.932
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     82/100      3.82G      1.173     0.9195      1.132         23        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.903       0.83      0.932
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     83/100      3.82G      1.209     0.9994      1.189         21        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.874      0.862      0.927
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     84/100      3.82G       1.29      1.056      1.167         19        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.891      0.871      0.938
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     85/100      3.82G      1.157      0.908      1.121         19        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.1it/s 0.1s
                   all         10         54      0.891      0.871      0.938
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     86/100      3.82G      1.185     0.9085      1.109         28        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.828      0.894      0.941
0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     87/100      3.82G      1.192     0.9501      1.115         14        640: 100%
```

```
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54       0.87      0.856      0.947
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     88/100      3.82G      1.174     0.9108       1.12         39        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54       0.87      0.856      0.947
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     89/100      3.82G      1.227     0.8994      1.173         16        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.882      0.867      0.947
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     90/100      3.82G      1.206     0.9226      1.156         15        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.882      0.867      0.947
0.58
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     91/100      3.82G      1.105     0.9094      1.152         16        640: 100%
━━━━━━━━━━━━ 5/5 1.7it/s 3.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.894      0.847      0.946
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     92/100      3.82G      1.135     0.8768       1.12         14        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 1.0s0.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.913      0.832      0.946
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     93/100      3.82G      1.128     0.8562      1.126         15        640: 100%
━━━━━━━━━━━━ 5/5 5.7it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.9it/s 0.1s
                   all         10         54      0.913      0.832      0.946
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
     94/100      3.82G       1.07     0.7945      1.121         14        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.899      0.839      0.939
0.581

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     95/100      3.82G      1.094     0.8064       1.11         15        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.893      0.845      0.934
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     96/100      3.82G      1.154     0.8627       1.14         16        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.1it/s 0.1s
                   all         10         54      0.893      0.845      0.934
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     97/100      3.82G      1.106     0.8054      1.134         15        640: 100%
━━━━━━━━━━━━ 5/5 5.3it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.4it/s 0.1s
                   all         10         54      0.874      0.862      0.936
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     98/100      3.82G      1.067     0.7839      1.081         16        640: 100%
━━━━━━━━━━━━ 5/5 5.6it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.874      0.862      0.936
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     99/100      3.82G      1.107     0.8389      1.109         16        640: 100%
━━━━━━━━━━━━ 5/5 5.4it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54       0.89      0.846      0.937
0.579

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
    100/100      3.82G      1.077     0.7797       1.05         16        640: 100%
━━━━━━━━━━━━ 5/5 5.5it/s 0.9s0.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.872      0.851      0.936
0.577

100 epochs completed in 0.076 hours.
```

```
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v1_baseline\weights\l
ast.pt, 52.0MB
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v1_baseline\weights\b
est.pt, 52.0MB

Validating D:\workplace_00\runs\detect\runs\mnist\v1_baseline\weights\best.pt...
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 3.3it/s 0.3s
                   all         10         54      0.773      0.936      0.964
0.597
                     0          4          6      0.702          1      0.995
0.717
                     1          3          3      0.748          1      0.995
0.601
                     2          7         10      0.889      0.798      0.917
0.482
                     3          6          8      0.803       0.75      0.916
0.519
                     4          4          4          1      0.999      0.995
0.59
                     5          1          1      0.311          1      0.995
0.697
                     6          3          5      0.802      0.814      0.962
0.607
                     7          3          3      0.873          1      0.995
0.588
                     8          5          7      0.675          1      0.871
0.475
                     9          6          7      0.933          1      0.995
0.693
Speed: 1.5ms preprocess, 23.2ms inference, 0.0ms loss, 0.9ms postprocess per image
Results saved to D:\workplace_00\runs\detect\runs\mnist\v1_baseline
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 130.543.3 MB/s, size: 5.9 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all         10         54      0.774      0.936      0.964
0.598
                     0          4          6      0.703          1      0.995
0.717
                     1          3          3      0.745          1      0.995
0.601
                     2          7         10      0.889      0.798      0.917
0.487
                     3          6          8      0.803       0.75      0.916
0.523
                     4          4          4          1      0.999      0.995
0.59
                     5          1          1      0.314          1      0.995
```

```
0.697
                     6          3          5      0.802      0.811      0.962
0.607
                     7          3          3      0.873          1      0.995
0.588
                     8          5          7      0.675          1      0.871
0.475
                     9          6          7      0.932          1      0.995
0.693
Speed: 1.0ms preprocess, 28.5ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val11

===== V1 No-Flip Validation Metrics =====
Precision   : 0.7736
Recall      : 0.9357
mAP50       : 0.9636
mAP50-95    : 0.5978
```

Result Interpretation

The baseline model achieved a reasonably strong initial performance, indicating that YOLO could learn the digit detection task even from a relatively small dataset. However, this experiment also served as a diagnostic stage, revealing that the default training configuration may include augmentations that are not fully suitable for handwritten digit detection.

10. Experiment 2: V1 Training Without Flip Augmentation

The second experiment reused **Version 1** of the dataset but changed the training policy by disabling flip augmentation.

This modification was introduced after recognizing that handwritten digits are fundamentally different from natural image objects. In natural image detection, horizontal flip is usually harmless or beneficial. In handwritten digit detection, however, a flipped digit may no longer represent a realistic or semantically correct example.

Therefore, this experiment was designed to test whether a more task-appropriate augmentation policy would improve detection quality even without changing the dataset version itself.

This experiment isolates the effect of the training policy while keeping the dataset fixed.

In [ ]:
```python
from ultralytics import YOLO

model_v1_noflip = YOLO("yolov8m.pt")

results_v1_noflip = model_v1_noflip.train(
    data="./data/mnist/mnist_v1/data_local.yaml",
```

```
    epochs=100,
    patience=50,
    imgsz=640,
    batch=16,
    device=0,
    fliplr=0.0,
    flipud=0.0,
    project="./runs/mnist",
    name="v1_no_flip",
    exist_ok=True
)

best_model = YOLO("./runs/detect/runs/mnist/v1_no_flip/weights/best.pt")

metrics = best_model.val(
    data="./data/mnist/mnist_v1/data_local.yaml",
    split="val",
    device=0
)

precision = metrics.box.mp
recall = metrics.box.mr
map50 = metrics.box.map50
map5095 = metrics.box.map

print("\n===== V1 No-Flip Validation Metrics =====")
print(f"Precision   : {precision:.4f}")
print(f"Recall      : {recall:.4f}")
print(f"mAP50       : {map50:.4f}")
print(f"mAP50-95    : {map5095:.4f}")
```

```
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment
=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close
_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=./data/mnist/mnist_v1/data_local.yaml, degrees=0.0, d
eterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=N
one, end2end=None, epochs=100, erasing=0.4, exist_ok=True, fliplr=0.0, flipud=0.0, f
ormat=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hs
v_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0
=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=v1_no_flip, nbs=64, nms=False, ops
et=None, optimize=False, optimizer=auto, overlap_mask=True, patience=50, perspective
=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=./runs/mnist, r
ect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, sa
ve_crop=False, save_dir=D:\workplace_00\runs\detect\runs\mnist\v1_no_flip, save_fram
es=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=
0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, s
ingle_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=Non
e, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=
0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module                                       a
rguments
  0                  -1  1      1392  ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]
  1                  -1  1     41664  ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]
  2                  -1  2    111360  ultralytics.nn.modules.block.C2f
[96, 96, 2, True]
  3                  -1  1    166272  ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]
  4                  -1  4    813312  ultralytics.nn.modules.block.C2f
[192, 192, 4, True]
  5                  -1  1    664320  ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]
  6                  -1  4   3248640  ultralytics.nn.modules.block.C2f
[384, 384, 4, True]
  7                  -1  1   1991808  ultralytics.nn.modules.conv.Conv
[384, 576, 3, 2]
  8                  -1  2   3985920  ultralytics.nn.modules.block.C2f
[576, 576, 2, True]
  9                  -1  1    831168  ultralytics.nn.modules.block.SPPF
[576, 576, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  2   1993728  ultralytics.nn.modules.block.C2f
[960, 384, 2]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
```

```
 15                  -1  2    517632  ultralytics.nn.modules.block.C2f
[576, 192, 2]
 16                  -1  1    332160  ultralytics.nn.modules.conv.Conv
[192, 192, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  2   1846272  ultralytics.nn.modules.block.C2f
[576, 384, 2]
 19                  -1  1   1327872  ultralytics.nn.modules.conv.Conv
[384, 384, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  2   4207104  ultralytics.nn.modules.block.C2f
[960, 576, 2]
 22        [15, 18, 21]  1   3781486  ultralytics.nn.modules.head.Detect
[10, 16, None, [192, 384, 576]]
Model summary: 170 layers, 25,862,110 parameters, 25,862,094 gradients, 79.1 GFLOPs

Transferred 469/475 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.00.0 ms, read: 18.412.0 MB/s, size: 4.9 KB)
train: Scanning D:\workplace_00\data\mnist\mnist_v1\train\labels.cache... 35 images,
0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 35/35  0.0s
val: Fast image access  (ping: 0.10.0 ms, read: 6.11.6 MB/s, size: 5.2 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
rmining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000714, momentum=0.9) with parameter groups 77 weight(decay=0.
0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Plotting labels to D:\workplace_00\runs\detect\runs\mnist\v1_no_flip\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to D:\workplace_00\runs\detect\runs\mnist\v1_no_flip
Starting training for 100 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      6.17G      2.595      4.703      1.876         11        640: 100%
━━━━━━━━━━━━ 3/3 1.4it/s 2.2s0.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 6.3it/s 0.2s
                   all         10         54     0.0107      0.282     0.0193    0.0
0435

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100      6.12G       2.72      4.822      1.816         20        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54     0.0122      0.189      0.022    0.0
0529

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100      6.11G      2.613      4.611      1.934         23        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54     0.0199      0.206     0.0403     0.
0103

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100      6.34G      2.128      3.936      1.541         26        640: 100%
━━━━━━━━━━━━ 3/3 2.7it/s 1.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54     0.0279      0.369     0.0528
0.016

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      6.39G      1.929      3.389      1.316         27        640: 100%
━━━━━━━━━━━━ 3/3 3.0it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54      0.136      0.121      0.119     0.
0375

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100       6.3G      1.649      2.911       1.32         16        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54      0.201       0.13     0.0758     0.
0298

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100      6.49G      1.524      2.546        1.2         30        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54     0.0946      0.374      0.122     0.
0362

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100      6.32G      1.495      2.525      1.194         33        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.105      0.356      0.136
0.043

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100      6.44G      1.636      2.512      1.302         28        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54     0.0883      0.483       0.19     0.
0699

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      6.33G      1.504      2.347      1.232         23        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.183       0.25      0.209     0.
0809

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100      6.51G      1.479      2.207      1.183         28        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.181       0.27      0.121     0.
0462

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     12/100      6.39G      1.683      2.562      1.351         11        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.1it/s 0.1s
                   all         10         54      0.221      0.291      0.127     0.
0692

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     13/100      6.48G      1.574      2.233      1.299         20        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.3it/s 0.1s
                   all         10         54      0.293      0.543      0.407
0.224

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     14/100      6.38G       1.54      2.036      1.244         16        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.4it/s 0.1s
                   all         10         54      0.397      0.722      0.469
0.231

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     15/100      6.45G      1.548      2.125      1.334         18        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.1it/s 0.1s
                   all         10         54      0.467      0.731      0.517
0.271

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     16/100      6.31G      1.527      2.027      1.266         38        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 1.0s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.647      0.538      0.631
0.334

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     17/100       6.4G      1.536      1.882      1.314         20        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.1it/s 0.1s
                   all         10         54       0.51      0.611      0.644
0.371

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     18/100      6.32G      1.536      1.871      1.331         24        640: 100%
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.525      0.628      0.609
0.378

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     19/100       6.3G      1.548      1.664      1.357         26        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.712      0.571      0.604
0.375

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     20/100      6.39G      1.362       1.66      1.271         25        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.643      0.596      0.654
0.436

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     21/100      6.29G      1.367      1.475      1.246         42        640: 100%
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54       0.47      0.738      0.641
0.429

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     22/100      6.33G      1.418       1.66      1.329         24        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 1.0s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.715      0.549      0.617
0.365

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     23/100       6.3G      1.414      1.557       1.27         33        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.532      0.687       0.75
0.435

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     24/100      6.38G      1.386      1.453      1.281         27        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.568      0.636      0.778
0.46

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     25/100      6.34G      1.384       1.59       1.34         11        640: 100%
━━━━━━━━━━━━ 3/3 2.8it/s 1.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.439      0.759       0.64
0.388

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     26/100      6.31G       1.35      1.352      1.303         18        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.4it/s 0.1s
                   all         10         54      0.789       0.68      0.719
0.42

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     27/100      6.34G      1.476      1.395      1.395         17        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54       0.63      0.634      0.676
0.354

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     28/100      6.33G      1.474      1.509      1.413         25        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.9it/s 0.1s
                   all         10         54       0.68      0.768      0.847
0.484

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     29/100      6.32G      1.397      1.376      1.366         13        640: 100%
━━━━━━━━━━━━ 3/3 3.0it/s 1.0s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54       0.68      0.768      0.847
0.484

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     30/100      6.44G      1.567      1.478      1.399         27        640: 100%
━━━━━━━━━━━━ 3/3 2.8it/s 1.1s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.837       0.73      0.909
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     31/100      6.42G       1.57      1.376      1.344         20        640: 100%
```

```
━━━━━━━━━━━━ 3/3 2.8it/s 1.1s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.818      0.694      0.865
0.529

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     32/100      6.51G      1.531      1.446        1.4         15        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.636      0.797      0.861
0.523

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     33/100      6.39G      1.495      1.352      1.415         14        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.636      0.797      0.861
0.523

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     34/100      6.48G      1.356      1.242      1.274         27        640: 100%
━━━━━━━━━━━━ 3/3 2.6it/s 1.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.832       0.72      0.907
0.548

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     35/100      6.41G      1.394      1.306      1.312         34        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.765      0.781      0.908
0.565

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     36/100      6.48G       1.45      1.231      1.259         32        640: 100%
━━━━━━━━━━━━ 3/3 2.7it/s 1.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.799      0.719      0.851
0.497

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     37/100      6.35G      1.324      1.255      1.317         17        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.799      0.719      0.851
0.497

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     38/100      6.43G       1.34        1.2      1.331         20        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.747      0.775      0.821
0.489

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     39/100      6.42G      1.344       1.23      1.319         20        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.6it/s 0.1s
                   all         10         54      0.743      0.838      0.908
0.534

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     40/100      6.45G      1.384      1.169      1.298         28        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.724      0.836      0.913
0.518

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     41/100      6.31G      1.349      1.212      1.301         13        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.724      0.836      0.913
0.518

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     42/100      6.48G       1.36      1.269      1.348         16        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.711      0.827       0.93
0.543

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     43/100       6.5G      1.242      1.035      1.199         30        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.742      0.828      0.915
0.56

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     44/100      6.42G      1.415      1.056      1.341         22        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.845        0.8       0.95
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     45/100      6.32G      1.358      1.174      1.271         11        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.845        0.8       0.95
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     46/100      6.46G      1.219      1.047      1.173         17        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54       0.76      0.884       0.95
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     47/100      6.49G      1.304       1.22      1.223         15        640: 100%
━━━━━━━━━━━━ 3/3 2.7it/s 1.1s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.815      0.883      0.918
0.551

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     48/100      6.41G      1.364      1.259      1.214         20        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.1it/s 0.1s
                   all         10         54      0.789      0.871      0.925
0.548

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     49/100      6.39G      1.326      1.172      1.214         12        640: 100%
━━━━━━━━━━━━ 3/3 2.8it/s 1.1s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.789      0.871      0.925
0.548

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     50/100      6.43G      1.327      1.145      1.267         34        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.807      0.795      0.907
0.525

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     51/100      6.48G      1.203     0.9923      1.182         22        640: 100%
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.5it/s 0.1s
                   all         10         54      0.819      0.823      0.922
0.52

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100      6.45G      1.211     0.9883      1.225         35        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.818      0.824      0.922
0.573

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100      6.34G       1.31      1.046      1.257         31        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.4it/s 0.1s
                   all         10         54      0.818      0.824      0.922
0.573

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100      6.49G      1.191     0.9064      1.189         25        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.696      0.915      0.865
0.547

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100       6.5G      1.199      0.972       1.17         26        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54       0.73      0.876      0.849
0.517

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100      6.41G      1.228     0.9361      1.203         38        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.783      0.868      0.922
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     57/100      6.34G      1.195     0.9758      1.221         35        640: 100%
━━━━━━━━━━━━ 3/3 3.4it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.783      0.868      0.922
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     58/100      6.51G      1.223     0.9116      1.178         29        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.866      0.852      0.949
0.576

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     59/100       6.4G      1.136     0.8375      1.125         24        640: 100%
```

```
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54       0.92      0.792      0.953
0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     60/100      6.45G      1.115     0.8418       1.15         34        640: 100%
━━━━━━━━━━━━ 3/3 2.7it/s 1.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.871      0.803      0.938
0.604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     61/100      6.38G      1.082     0.8421      1.104         39        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.871      0.803      0.938
0.604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     62/100      6.45G      1.089     0.8928      1.198         23        640: 100%
━━━━━━━━━━━━ 3/3 2.8it/s 1.1s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.869      0.796      0.959
0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     63/100      6.47G      1.027     0.8468      1.164         12        640: 100%
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.781        0.9       0.94
0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     64/100      6.44G      1.395     0.9044       1.26         20        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.1it/s 0.1s
                   all         10         54      0.825      0.841      0.944
0.588

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     65/100      6.38G      1.241      0.889      1.155         25        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.825      0.841      0.944
0.588

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     66/100      6.44G      1.148     0.8927      1.164         25        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.7it/s 0.1s
                   all         10         54       0.88      0.858      0.958
0.51

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     67/100      6.39G      1.302      1.031      1.305         16        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.872      0.817      0.926
0.488

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     68/100      6.44G      1.107     0.8169      1.137         22        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.5it/s 0.1s
                   all         10         54      0.865      0.888      0.972
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     69/100      6.35G      1.052     0.7896      1.085         32        640: 100%
━━━━━━━━━━━━ 3/3 3.4it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.3it/s 0.1s
                   all         10         54      0.865      0.888      0.972
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     70/100      6.49G      1.105     0.7776      1.094         22        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.854      0.895      0.977
0.588

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     71/100      6.47G      1.134     0.8773      1.121         16        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.3it/s 0.1s
                   all         10         54      0.869      0.874      0.973
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     72/100      6.49G      1.158     0.8343      1.188         31        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.1it/s 0.1s
                   all         10         54      0.847      0.855      0.973
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     73/100      6.34G      1.025     0.7439      1.117         30        640: 100%
```

```
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.847      0.855      0.973
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     74/100      6.42G      1.174     0.8524      1.179         20        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.801      0.919      0.965
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     75/100      6.44G      1.055     0.8096      1.097         26        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.808      0.916      0.959
0.544

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     76/100      6.44G      1.074     0.9273      1.181         15        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54        0.9      0.813      0.959
0.539

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     77/100      6.32G      1.178     0.9106       1.26         14        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.3it/s 0.1s
                   all         10         54        0.9      0.813      0.959
0.539

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     78/100      6.44G      1.089     0.8541      1.109         28        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.5it/s 0.1s
                   all         10         54      0.888      0.814      0.964
0.581

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     79/100      6.43G      1.054     0.7924      1.132         21        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.872      0.847      0.966
0.605

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     80/100       6.4G      1.096     0.7862      1.083         20        640: 100%
```

```
━━━━━━━━━━━━ 3/3 2.9it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.841      0.872      0.969
0.594

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     81/100      6.34G      1.018     0.7241      1.123         22        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.841      0.872      0.969
0.594

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     82/100      6.49G      1.009     0.7151      1.067         29        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.3it/s 0.1s
                   all         10         54      0.868      0.868       0.98
0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     83/100      6.48G      1.074      0.809      1.086         14        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.894      0.848      0.976
0.566

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     84/100      6.44G      1.057     0.8423      1.128         18        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.904      0.845      0.977
0.568

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     85/100      6.38G      1.139     0.7812      1.173         15        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.904      0.845      0.977
0.568

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     86/100       6.5G      1.215     0.9179      1.284         14        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.8it/s 0.1s
                   all         10         54      0.898      0.837      0.975
0.585

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     87/100      6.48G      1.067     0.7754       1.09         20        640: 100%
```

```
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.886      0.892      0.979
0.599

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     88/100      6.46G     0.9259     0.6978      1.047         37        640: 100%
━━━━━━━━━━━━ 3/3 3.1it/s 1.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.868      0.885      0.975
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     89/100      6.38G      1.082       0.72      1.094         41        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.868      0.885      0.975
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     90/100      6.43G      1.076     0.7456      1.077         32        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.883      0.872      0.977
0.579
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     91/100       6.4G     0.8661     0.6709      1.052         14        640: 100%
━━━━━━━━━━━━ 3/3 1.4it/s 2.1s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.868      0.884      0.977
0.573

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     92/100      6.41G     0.9583     0.6522      1.082         17        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.903      0.838      0.978
0.568

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     93/100       6.3G     0.9468     0.6476      1.054         15        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.903      0.838      0.978
0.568

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
     94/100      6.4G     0.9114     0.6635      1.077         15        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.899      0.839      0.977
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     95/100     6.39G     0.9083     0.6404      1.058         14        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.8it/s 0.1s
                   all         10         54      0.868      0.875      0.976
0.559

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     96/100     6.41G     0.8224     0.5948      1.013         16        640: 100%
━━━━━━━━━━━━ 3/3 3.2it/s 0.9s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.866      0.884      0.976
0.553

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     97/100      6.3G     0.8996     0.6492      1.087         16        640: 100%
━━━━━━━━━━━━ 3/3 3.4it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.866      0.884      0.976
0.553

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     98/100     6.41G     0.8643       0.61      1.027         15        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.866      0.886      0.978
0.563

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     99/100      6.4G     0.9271     0.6619      1.089         20        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s2.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.866      0.883      0.976
0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
    100/100     6.41G     0.9003     0.6789      1.123         14        640: 100%
━━━━━━━━━━━━ 3/3 3.3it/s 0.9s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.868      0.886      0.974
0.569

100 epochs completed in 0.073 hours.
```

```
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v1_no_flip\weights\la
st.pt, 52.0MB
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v1_no_flip\weights\be
st.pt, 52.0MB

Validating D:\workplace_00\runs\detect\runs\mnist\v1_no_flip\weights\best.pt...
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.873      0.847      0.966
0.605
                     0          4          6      0.855          1      0.995
0.699
                     1          3          3      0.733          1      0.913
0.596
                     2          7         10          1      0.571      0.972
0.499
                     3          6          8      0.889        0.5      0.923
0.539
                     4          4          4          1      0.877      0.995
0.564
                     5          1          1      0.481          1      0.995
0.697
                     6          3          5      0.796        0.8       0.88
0.542
                     7          3          3      0.975          1      0.995
0.559
                     8          5          7          1      0.757      0.995
0.622
                     9          6          7          1      0.964      0.995
0.736
Speed: 1.2ms preprocess, 7.0ms inference, 0.0ms loss, 0.6ms postprocess per image
Results saved to D:\workplace_00\runs\detect\runs\mnist\v1_no_flip
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 119.438.1 MB/s, size: 5.7 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.6s/it 1.6s
                   all         10         54      0.872      0.847      0.966
0.6
                     0          4          6      0.854          1      0.995
0.699
                     1          3          3      0.733          1      0.913
0.554
                     2          7         10          1      0.572      0.972
0.483
                     3          6          8      0.889        0.5      0.923
0.548
                     4          4          4          1      0.877      0.995
0.564
                     5          1          1      0.466          1      0.995
```

```
0.697
                     6         3         5     0.799       0.8      0.88
0.542
                     7         3         3     0.974         1     0.995
0.559
                     8         5         7         1     0.758     0.995
0.622
                     9         6         7         1     0.964     0.995
0.736
Speed: 2.2ms preprocess, 43.6ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val8

===== V1 No-Flip Validation Metrics =====
Precision   : 0.8715
Recall      : 0.8472
mAP50       : 0.9658
mAP50-95    : 0.6004
```

Result Interpretation

Compared with the baseline model, the no-flip version improved precision, mAP50, and mAP50-95, while recall decreased. This indicates that removing horizontal flip augmentation made the detector more conservative but also more reliable, which is consistent with the idea that flip transformations are not always appropriate for handwritten digit detection.

11. Experiment 3: Training on Dataset Version 2

The third experiment used **Version 2** of the dataset. Compared with Version 1, Version 2 included additional augmentation settings applied in Roboflow and contained a larger number of training images.

Unlike the previous experiment, the purpose here was not to change the training policy, but to evaluate whether improving the dataset itself could produce better results. To make the comparison fair, the no-flip training setting was kept consistent with the improved V1 experiment.

This experiment therefore focuses on the effect of **dataset diversity and augmentation-managed versioning**.

In [ ]:

```python
from ultralytics import YOLO

model_v2_noflip = YOLO("yolov8m.pt")

results_v2_noflip = model_v2_noflip.train(
    data="./data/mnist/mnist_v2/data_local.yaml",
    epochs=100,
    patience=50,
    imgsz=640,
    batch=16,
    device=0,
```

```
    fliplr=0.0,
    flipud=0.0,
    project="./runs/mnist",
    name="v2_no_flip",
    exist_ok=True
)

best_model = YOLO("./runs/detect/runs/mnist/v2_no_flip/weights/best.pt")

metrics = best_model.val(
    data="./data/mnist/mnist_v2/data_local.yaml",
    split="val",
    device=0
)

precision = metrics.box.mp
recall = metrics.box.mr
map50 = metrics.box.map50
map5095 = metrics.box.map

print("\n===== V2 No-Flip Validation Metrics =====")
print(f"Precision    : {precision:.4f}")
print(f"Recall       : {recall:.4f}")
print(f"mAP50        : {map50:.4f}")
print(f"mAP50-95     : {map5095:.4f}")
```

```
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment
=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close
_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=./data/mnist/mnist_v2/data_local.yaml, degrees=0.0, d
eterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=N
one, end2end=None, epochs=100, erasing=0.4, exist_ok=True, fliplr=0.0, flipud=0.0, f
ormat=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hs
v_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0
=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=v2_no_flip, nbs=64, nms=False, ops
et=None, optimize=False, optimizer=auto, overlap_mask=True, patience=50, perspective
=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=./runs/mnist, r
ect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, sa
ve_crop=False, save_dir=D:\workplace_00\runs\detect\runs\mnist\v2_no_flip, save_fram
es=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=
0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, s
ingle_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=Non
e, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=
0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module                                       a
rguments
  0                  -1  1      1392  ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]
  1                  -1  1     41664  ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]
  2                  -1  2    111360  ultralytics.nn.modules.block.C2f
[96, 96, 2, True]
  3                  -1  1    166272  ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]
  4                  -1  4    813312  ultralytics.nn.modules.block.C2f
[192, 192, 4, True]
  5                  -1  1    664320  ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]
  6                  -1  4   3248640  ultralytics.nn.modules.block.C2f
[384, 384, 4, True]
  7                  -1  1   1991808  ultralytics.nn.modules.conv.Conv
[384, 576, 3, 2]
  8                  -1  2   3985920  ultralytics.nn.modules.block.C2f
[576, 576, 2, True]
  9                  -1  1    831168  ultralytics.nn.modules.block.SPPF
[576, 576, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  2   1993728  ultralytics.nn.modules.block.C2f
[960, 384, 2]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
```

```
 15                  -1  2    517632  ultralytics.nn.modules.block.C2f
[576, 192, 2]
 16                  -1  1    332160  ultralytics.nn.modules.conv.Conv
[192, 192, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  2   1846272  ultralytics.nn.modules.block.C2f
[576, 384, 2]
 19                  -1  1   1327872  ultralytics.nn.modules.conv.Conv
[384, 384, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  2   4207104  ultralytics.nn.modules.block.C2f
[960, 576, 2]
 22        [15, 18, 21]  1   3781486  ultralytics.nn.modules.head.Detect
[10, 16, None, [192, 384, 576]]
Model summary: 170 layers, 25,862,110 parameters, 25,862,094 gradients, 79.1 GFLOPs

Transferred 469/475 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.10.0 ms, read: 12.413.6 MB/s, size: 11.3 KB)
train: Scanning D:\workplace_00\data\mnist\mnist_v2\train\labels.cache... 104 image
s, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 104/104  0.0s
val: Fast image access  (ping: 0.00.0 ms, read: 13.03.1 MB/s, size: 10.4 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v2\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
rmining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000714, momentum=0.9) with parameter groups 77 weight(decay=0.
0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Plotting labels to D:\workplace_00\runs\detect\runs\mnist\v2_no_flip\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to D:\workplace_00\runs\detect\runs\mnist\v2_no_flip
Starting training for 100 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      6.64G      2.858       4.68       2.21         40        640: 100%
━━━━━━━━━━━━ 7/7 1.3it/s 5.2s0.7ss
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 3.9it/s 0.3s
                   all         10         54      0.109      0.025     0.0161      0.0
0453

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100      6.62G      2.191        3.8       1.66         45        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.139      0.245      0.159       0.
0785

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100      6.81G      1.608      2.711      1.288         65        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.147      0.322      0.184     0.
0837

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100      6.82G      1.486      2.253       1.23         50        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.317      0.488      0.401
0.226

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      7.01G      1.434      2.111      1.235         39        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.495      0.539      0.586
0.318

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100      6.85G       1.43      1.849      1.281         43        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.6it/s 0.1s
                   all         10         54      0.613      0.636        0.7
0.429

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100      6.93G      1.465      1.798       1.34         26        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.597      0.697      0.764
0.438

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100      6.88G      1.402      1.587      1.236         40        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.749      0.661      0.781
0.425

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100       6.9G      1.397      1.507      1.277         52        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.3it/s 0.1s
                   all         10         54      0.746      0.782      0.784
0.439

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      6.85G      1.415      1.509      1.273         53        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54      0.607      0.856      0.749
0.426

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100         7G      1.324      1.373      1.228         83        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.662      0.738       0.72
0.374

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     12/100      6.99G      1.382      1.265      1.233         60        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.605      0.582      0.665
0.319

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     13/100      6.82G       1.43       1.37       1.29         75        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.661      0.638      0.681
0.371

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     14/100       6.9G       1.35        1.3      1.207         71        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.6it/s 0.1s
                   all         10         54      0.706      0.663      0.702
0.365

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     15/100      6.98G      1.426      1.298      1.256         75        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.851      0.706      0.764
0.399

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     16/100      6.93G       1.29       1.11      1.202         65        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.733      0.811      0.887
0.502

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     17/100      6.89G      1.333      1.164      1.204         78        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.736      0.673      0.741
0.364

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     18/100      6.95G      1.283      1.073      1.197         59        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.782      0.727      0.843
0.396

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     19/100      6.91G      1.266      1.093      1.177         80        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54      0.772      0.757      0.888
0.496

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     20/100      6.93G       1.31      1.113      1.187         67        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.781      0.774      0.854
0.458

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     21/100       6.8G      1.316       1.15        1.2         47        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.6it/s 0.1s
                   all         10         54      0.731      0.859      0.892
0.552

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     22/100      6.92G      1.364      1.112      1.237         46        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.2it/s 0.1s
                   all         10         54       0.78      0.853      0.915
0.51

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     23/100      6.98G      1.306      1.094      1.208         68        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.808      0.829      0.947
0.585

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     24/100      6.93G      1.255     0.9788      1.173         70        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.5it/s 2.9s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.783      0.792      0.908
0.511

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     25/100       6.8G      1.212      1.022      1.176         41        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.9it/s 0.1s
                   all         10         54      0.802      0.828      0.908
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     26/100      6.91G      1.199     0.9024      1.153         73        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.871      0.873      0.949
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     27/100      6.92G      1.208     0.9489      1.164         55        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.912      0.795      0.943
0.546

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     28/100      6.99G       1.19     0.8968      1.178         42        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.903      0.866      0.957
0.562

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     29/100      6.82G      1.195     0.8642      1.156         61        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.875      0.862      0.929
0.526

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     30/100      6.92G      1.163      0.839      1.148         54        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.788      0.861      0.935
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     31/100      6.94G      1.114     0.7956      1.123         58        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.4it/s 0.1s
                   all         10         54      0.825      0.825       0.92
0.563

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     32/100      6.92G      1.158     0.8627      1.149         66        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.861      0.836      0.925
0.519

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     33/100      6.81G      1.156     0.8122      1.149         59        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.835      0.793      0.918
0.482

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     34/100      6.94G      1.127     0.7688      1.128         68        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.828      0.839       0.94
0.515

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     35/100      6.95G      1.148     0.8409      1.142         47        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.813       0.86      0.937
0.504

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     36/100      6.96G      1.148     0.8331       1.18         50        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.755      0.924      0.958
0.512

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     37/100      6.85G      1.135     0.8387      1.156         68        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.4it/s 0.1s
                   all         10         54      0.879      0.769      0.936
0.492

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     38/100      6.93G      1.083     0.7875      1.108         76        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.771      0.756      0.804
0.407

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     39/100      6.94G      1.128     0.7988      1.144         51        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.771       0.81      0.857
0.454

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     40/100      6.95G      1.044     0.7323      1.092         59        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54       0.84      0.777      0.851
0.47

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     41/100      6.85G      1.106     0.7407      1.111         66        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.867      0.853      0.949
0.531

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     42/100      6.92G      1.044     0.7383      1.087         68        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.819      0.853       0.91
0.502

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     43/100      6.92G     0.9961     0.6806      1.058         59        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.912      0.839      0.968
0.518

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     44/100      6.92G      1.007     0.6757      1.086         41        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.839      0.921      0.964
0.55

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     45/100      6.85G      1.017     0.6903      1.075         74        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.843      0.906      0.966
0.548

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     46/100      6.98G     0.9996     0.6793      1.052         55        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.907      0.883      0.969
0.525

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     47/100      6.99G      1.091     0.6974       1.12         53        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54       0.91      0.857      0.952
0.506

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     48/100      6.92G      1.033     0.7173      1.083         60        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.924      0.862      0.969
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     49/100      6.88G      0.971     0.6837      1.101         53        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54       0.93      0.843      0.963
0.569

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     50/100      6.98G     0.9354     0.6543      1.058         48        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.886      0.856       0.96
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     51/100      6.91G     0.9251     0.6444      1.041         72        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.844      0.879      0.953
0.561

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100      6.93G     0.9329     0.6466      1.026         57        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.863      0.868      0.954
0.515

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100      6.89G     0.9845     0.6561      1.067         48        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.851       0.86      0.961
0.52

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100      6.94G     0.9111     0.6097      1.007         71        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.854      0.852      0.952
0.522

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100         7G      0.951     0.6272      1.035         58        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54       0.85      0.825      0.902
0.526

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100      6.93G     0.8919     0.6024      1.019         57        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54        0.9       0.87      0.963
0.55

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     57/100      6.85G     0.9232     0.6135      1.022         45        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.845       0.86      0.948
0.552

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     58/100      6.95G     0.8227     0.5762     0.9988         90        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.859       0.84       0.95
0.559

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     59/100       6.9G     0.8667     0.5932       1.02         47        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.899      0.926      0.966
0.537

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     60/100      6.93G     0.8254     0.5742     0.9983         47        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.891       0.93      0.956
0.561

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     61/100      6.82G     0.8668     0.5597      1.014         50        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.5it/s 0.1s
                   all         10         54       0.89      0.923      0.957
0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     62/100      6.91G      0.883     0.5694      1.011         60        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.845      0.909      0.931
0.513

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     63/100      6.91G     0.9627     0.5953      1.049         69        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.841      0.931      0.958
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     64/100      6.99G     0.9281     0.6023      1.055         41        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.835        0.9       0.94
0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     65/100      6.81G     0.9101     0.6008      1.025         84        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.844      0.888      0.937
0.546

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     66/100      6.94G     0.9101     0.5883      1.042         86        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.4it/s 0.1s
                   all         10         54      0.783      0.907      0.927
0.492

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     67/100      6.98G     0.8414     0.5521      1.001         57        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.823      0.913      0.962
0.568

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     68/100      6.91G     0.8647     0.5559      1.015         30        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.832      0.872      0.947
0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     69/100      6.81G     0.8042     0.5313     0.9854         55        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54       0.84      0.871      0.954
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     70/100      6.92G     0.8094     0.5583     0.9855         51        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.838      0.953      0.976
0.579

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     71/100      6.92G     0.7827     0.5389     0.9851         65        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.864      0.931       0.97
0.578

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     72/100         7G     0.7633     0.4911     0.9579         69        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54       0.85      0.898       0.97
0.595

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     73/100      6.86G     0.7685     0.4889     0.9549         39        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54       0.854      0.897       0.96
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     74/100      6.92G     0.7396     0.4928      0.955         63        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54       0.867      0.844      0.936
0.556

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     75/100      6.92G     0.7814     0.5207     0.9655        101        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54       0.847      0.846      0.928
0.551

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     76/100      6.96G     0.7095     0.4929     0.9494         78        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54       0.844      0.896      0.949
0.563

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     77/100      6.81G     0.7492     0.4918     0.9497         65        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54       0.871       0.87      0.952
0.557

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     78/100         7G     0.7881     0.5283     0.9867         44        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54       0.858      0.891      0.953
0.543

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     79/100      6.97G     0.6608     0.4594     0.9199         68        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.6it/s 0.1s
                   all         10         54       0.842      0.899      0.945
0.542

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     80/100      6.95G     0.6909     0.4764      0.931         52        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54       0.85      0.924      0.952
0.534

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     81/100      6.82G     0.6899     0.4749     0.9401         60        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.9it/s 0.1s
                   all         10         54      0.847      0.921      0.953
0.54

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     82/100      6.98G     0.7071     0.4797     0.9488         73        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54      0.832      0.938      0.952
0.538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     83/100       6.9G     0.6863     0.4672     0.9479         39        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.3it/s 0.1s
                   all         10         54      0.861      0.894      0.954
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     84/100      6.96G     0.7352     0.5042     0.9482         71        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.882      0.875      0.962
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     85/100      6.83G     0.6457     0.4628     0.9232         63        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.879      0.884      0.969
0.557

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     86/100       6.9G     0.7232     0.4855     0.9573         45        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.1it/s 0.1s
                   all         10         54      0.885      0.884      0.973
0.554

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     87/100      6.97G     0.6681     0.4599     0.9428         59        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.1it/s 0.1s
                   all         10         54      0.864      0.916      0.976
0.564

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     88/100      6.95G     0.7111     0.4626     0.9617         35        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.8it/s 0.1s
                   all         10         54      0.864      0.918       0.97
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     89/100       6.9G     0.7166     0.4699     0.9314         44        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.887      0.889       0.97
0.564

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     90/100         7G     0.6686     0.4654     0.9293         41        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.868      0.863      0.946
0.553
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     91/100      6.91G     0.7063     0.4118     0.9458         40        640: 100%
━━━━━━━━━━━━ 7/7 1.4it/s 5.0s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.896      0.881       0.97
0.549

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     92/100      6.91G     0.5838      0.392      0.913         28        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.886      0.898       0.97
0.548

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     93/100       6.8G     0.6735     0.3947     0.9222         48        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.6it/s 0.1s
                   all         10         54      0.887      0.918      0.977
0.557

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
     94/100      6.9G     0.6981     0.4221     0.9511         39        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.888      0.919      0.976
0.561

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     95/100      6.9G     0.5672     0.3764     0.8815         46        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54       0.89      0.905      0.975
0.565

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     96/100     6.91G     0.6114     0.4191     0.9335         26        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.874      0.879      0.948
0.552

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     97/100     6.81G     0.5989     0.3882     0.9232         39        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.873      0.879      0.948
0.555

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     98/100     6.89G     0.5763       0.35     0.8917         40        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.869      0.881      0.948
0.559

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     99/100     6.91G     0.5649     0.3806     0.9044         40        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.0it/s 0.1s
                   all         10         54      0.863      0.885      0.947
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
    100/100     6.91G     0.5244     0.3455     0.8639         33        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.2it/s 0.1s
                   all         10         54      0.862      0.891      0.949
0.572

100 epochs completed in 0.116 hours.
```

```
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v2_no_flip\weights\la
st.pt, 52.0MB
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v2_no_flip\weights\be
st.pt, 52.0MB

Validating D:\workplace_00\runs\detect\runs\mnist\v2_no_flip\weights\best.pt...
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 3.6it/s 0.3s
                   all         10         54       0.85      0.898       0.97
0.595
                     0          4          6      0.799          1      0.995
0.66
                     1          3          3       0.86          1      0.995
0.404
                     2          7         10          1      0.862      0.962
0.583
                     3          6          8          1      0.841      0.982
0.644
                     4          4          4      0.725      0.674      0.895
0.595
                     5          1          1      0.423          1      0.995
0.796
                     6          3          5      0.905        0.8      0.938
0.487
                     7          3          3          1      0.942      0.995
0.492
                     8          5          7      0.986          1      0.995
0.579
                     9          6          7      0.806      0.857      0.944
0.711
Speed: 0.2ms preprocess, 17.1ms inference, 0.0ms loss, 0.7ms postprocess per image
Results saved to D:\workplace_00\runs\detect\runs\mnist\v2_no_flip
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 190.146.4 MB/s, size: 10.0 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v2\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all         10         54       0.85      0.898       0.97
0.588
                     0          4          6        0.8          1      0.995
0.66
                     1          3          3       0.86          1      0.995
0.404
                     2          7         10          1      0.862      0.962
0.583
                     3          6          8          1      0.841      0.982
0.644
                     4          4          4      0.725      0.674      0.895
0.595
                     5          1          1      0.423          1      0.995
```

```
0.796
                     6          3          5     0.905       0.8     0.938
0.47
                     7          3          3         1     0.941     0.995
0.449
                     8          5          7     0.986         1     0.995
0.572
                     9          6          7     0.806     0.857     0.944
0.711
Speed: 5.4ms preprocess, 40.2ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val9

===== V2 No-Flip Validation Metrics =====
Precision   : 0.8504
Recall      : 0.8976
mAP50       : 0.9695
mAP50-95    : 0.5882
```

Result Interpretation

The Version 2 dataset improved recall and achieved the highest mAP50 among all experiments, indicating stronger object coverage and better overall detection performance at the standard IoU threshold. However, its mAP50-95 was lower than that of the best final model, suggesting that stricter localization quality still had room for improvement.

12. Experiment 4: Training on Dataset Version 3

The final experiment used **Version 3**, the most refined dataset version prepared in this study. This version was trained under the same no-flip condition used for the previous improved experiments.

The purpose of this final stage was to determine whether an additional dataset refinement step could further improve performance and produce the most balanced detector among all experimental settings.

Because the model architecture and training policy remained consistent, the performance of Version 3 can be interpreted primarily as the result of dataset refinement.

In [ ]:

```
from ultralytics import YOLO

model_v3_noflip = YOLO("yolov8m.pt")

results_v3_noflip = model_v3_noflip.train(
    data="./data/mnist/mnist_v3/data_local.yaml",
    epochs=100,
    patience=50,
    imgsz=640,
    batch=16,
    device=0,
    fliplr=0.0,
```

```python
    flipud=0.0,
    project="./runs/mnist",
    name="v3_no_flip",
    exist_ok=True
)

best_model = YOLO("./runs/detect/runs/mnist/v3_no_flip/weights/best.pt")

metrics = best_model.val(
    data="./data/mnist/mnist_v3/data_local.yaml",
    split="val",
    device=0
)

precision = metrics.box.mp
recall = metrics.box.mr
map50 = metrics.box.map50
map5095 = metrics.box.map

print("\n===== V3 No-Flip Validation Metrics =====")
print(f"Precision   : {precision:.4f}")
print(f"Recall      : {recall:.4f}")
print(f"mAP50       : {map50:.4f}")
print(f"mAP50-95    : {map5095:.4f}")
```

```
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment
=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close
_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=./data/mnist/mnist_v3/data_local.yaml, degrees=0.0, d
eterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=N
one, end2end=None, epochs=100, erasing=0.4, exist_ok=True, fliplr=0.0, flipud=0.0, f
ormat=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hs
v_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0
=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=v3_no_flip, nbs=64, nms=False, ops
et=None, optimize=False, optimizer=auto, overlap_mask=True, patience=50, perspective
=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=./runs/mnist, r
ect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, sa
ve_crop=False, save_dir=D:\workplace_00\runs\detect\runs\mnist\v3_no_flip, save_fram
es=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=
0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, s
ingle_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=Non
e, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=
0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module                                       a
rguments
  0                  -1  1      1392  ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]
  1                  -1  1     41664  ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]
  2                  -1  2    111360  ultralytics.nn.modules.block.C2f
[96, 96, 2, True]
  3                  -1  1    166272  ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]
  4                  -1  4    813312  ultralytics.nn.modules.block.C2f
[192, 192, 4, True]
  5                  -1  1    664320  ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]
  6                  -1  4   3248640  ultralytics.nn.modules.block.C2f
[384, 384, 4, True]
  7                  -1  1   1991808  ultralytics.nn.modules.conv.Conv
[384, 576, 3, 2]
  8                  -1  2   3985920  ultralytics.nn.modules.block.C2f
[576, 576, 2, True]
  9                  -1  1    831168  ultralytics.nn.modules.block.SPPF
[576, 576, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  2   1993728  ultralytics.nn.modules.block.C2f
[960, 384, 2]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
```

```
 15                  -1  2    517632  ultralytics.nn.modules.block.C2f
[576, 192, 2]
 16                  -1  1    332160  ultralytics.nn.modules.conv.Conv
[192, 192, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  2   1846272  ultralytics.nn.modules.block.C2f
[576, 384, 2]
 19                  -1  1   1327872  ultralytics.nn.modules.conv.Conv
[384, 384, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  2   4207104  ultralytics.nn.modules.block.C2f
[960, 576, 2]
 22        [15, 18, 21]  1   3781486  ultralytics.nn.modules.head.Detect
[10, 16, None, [192, 384, 576]]
Model summary: 170 layers, 25,862,110 parameters, 25,862,094 gradients, 79.1 GFLOPs

Transferred 469/475 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.00.0 ms, read: 20.618.4 MB/s, size: 11.3 KB)
train: Scanning D:\workplace_00\data\mnist\mnist_v3\train\labels.cache... 105 image
s, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 105/105  0.0s
val: Fast image access  (ping: 0.10.0 ms, read: 23.716.1 MB/s, size: 10.4 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v3\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
rmining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000714, momentum=0.9) with parameter groups 77 weight(decay=0.
0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Plotting labels to D:\workplace_00\runs\detect\runs\mnist\v3_no_flip\labels.jpg...
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to D:\workplace_00\runs\detect\runs\mnist\v3_no_flip
Starting training for 100 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      6.15G      3.032      4.948      2.413         44        640: 100%
━━━━━━━━━━━━ 7/7 1.3it/s 5.2s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 3.0it/s 0.3s
                   all         10         54      0.108      0.025     0.0158    0.0
0442

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100      6.08G      2.293      3.952      1.735         55        640: 100%
━━━━━━━━━━━━ 7/7 2.4it/s 2.9s0.4ss
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.1it/s 0.1s
                   all         10         54      0.115      0.176       0.12
0.053

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100      6.38G      1.657       2.81      1.327         81        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.229      0.329      0.168     0.
0934

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100      6.34G      1.483      2.384      1.207         64        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.188      0.582        0.2
0.111

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      6.45G      1.531      2.279       1.28         58        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.264      0.712      0.407
0.242

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100      6.32G      1.451      2.069      1.238         47        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.8it/s 0.1s
                   all         10         54      0.545        0.7       0.68
0.382

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100      6.43G      1.481      1.818      1.307         36        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54       0.56      0.607      0.674
0.399

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100      6.37G      1.412      1.676      1.252         60        640: 100%
━━━━━━━━━━━━ 7/7 2.4it/s 2.9s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54       0.56      0.787       0.74
0.466

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100      6.48G      1.385      1.554      1.244         61        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.769      0.664      0.787
0.478

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      6.31G      1.423      1.592      1.271         64        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.718      0.857      0.863
0.512

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100      6.44G      1.415      1.407      1.251         87        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.605       0.83      0.829
0.429

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     12/100      6.42G      1.344      1.313      1.201         81        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54       0.72      0.747      0.865
0.479

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     13/100      6.34G      1.523      1.422      1.332         60        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.784      0.733      0.873
0.463

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     14/100      6.42G      1.433      1.296      1.259         69        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.6it/s 0.1s
                   all         10         54      0.771      0.778      0.873
0.46

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     15/100      6.41G      1.465      1.317      1.247         81        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.489      0.465      0.475
0.181

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     16/100      6.42G       1.37      1.275      1.225         58        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.547      0.383      0.387
0.154

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     17/100      6.38G      1.328      1.215        1.2         88        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.617      0.279      0.317     0.
0846

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     18/100      6.44G       1.32      1.197      1.172         72        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.803      0.818      0.852
0.401

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     19/100      6.41G      1.372      1.263      1.182         87        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.773      0.862      0.943
0.551

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     20/100      6.51G      1.323      1.124      1.151         82        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.914      0.844      0.956
0.51

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     21/100      6.38G      1.336      1.148      1.184         62        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54       0.82      0.917      0.948
0.546

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     22/100      6.47G      1.416      1.125      1.244         48        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.827      0.917      0.963
0.57

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     23/100      6.43G      1.327      1.084      1.206         77        640: 100%
━━━━━━━━━━━━ 7/7 2.4it/s 2.9s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.942      0.874       0.97
0.506

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     24/100      6.41G       1.29      1.072      1.174        106        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.857       0.79      0.918
0.534

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     25/100      6.35G      1.305      1.012      1.174         61        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.865      0.731      0.859
0.384

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     26/100      6.38G      1.258     0.9923      1.171         71        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.875      0.882      0.939
0.544

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     27/100      6.48G      1.273     0.9853      1.158         80        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.853      0.877      0.924
0.527

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     28/100      6.49G      1.174     0.9595       1.13         50        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.946      0.838      0.969
0.56

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     29/100      6.32G      1.211     0.9322      1.138        100        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.793      0.887      0.963
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     30/100      6.49G      1.167     0.9004      1.138         44        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.883       0.78      0.944
0.533

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     31/100      6.44G       1.15     0.8017      1.104         78        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.857      0.832       0.91
0.391

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     32/100      6.41G      1.179     0.8611      1.144         60        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54        0.9      0.886       0.96
0.562

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     33/100      6.34G      1.256     0.9018      1.185         59        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.793      0.931      0.962
0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     34/100      6.44G      1.168     0.8318      1.123         55        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.791      0.859      0.923
0.48

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     35/100      6.47G       1.14     0.8336      1.114         79        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.874      0.869      0.949
0.531

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     36/100      6.44G      1.199     0.8705      1.162         63        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.749      0.918      0.911
0.514

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     37/100      6.31G      1.061     0.7649      1.091         76        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.763      0.938      0.949
0.539

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     38/100      6.42G       1.17     0.8086       1.13         45        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.814       0.91      0.952
0.543

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     39/100      6.48G       1.13     0.8471      1.114         55        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.792      0.852      0.928
0.547

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     40/100       6.4G      1.078     0.7971      1.065         80        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.0it/s 0.1s
                   all         10         54      0.708      0.829      0.876
0.367

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     41/100       6.3G      1.141     0.8157        1.1         81        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54       0.82      0.796      0.892
0.422

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     42/100      6.41G      1.164     0.8016      1.107         90        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.884      0.805      0.945
0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     43/100       6.5G      1.073     0.7483      1.081         62        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.5it/s 0.1s
                   all         10         54      0.875      0.874      0.945
0.538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     44/100      6.48G      1.062     0.7246      1.079         76        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.868      0.876      0.963
0.606

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     45/100      6.31G      1.052     0.7108      1.047         87        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.843      0.866      0.945
0.545

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     46/100      6.41G      1.047     0.7327      1.061         63        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 11.2it/s 0.1s
                   all         10         54       0.85      0.862      0.948
0.585

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     47/100      6.39G      1.068     0.7299      1.076         42        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.859      0.888       0.95
0.592

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     48/100      6.48G      1.024     0.7122      1.068         90        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.2it/s 0.1s
                   all         10         54      0.888      0.891      0.972
0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     49/100       6.4G      1.081     0.7027      1.076         70        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.889        0.9      0.966
0.603

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     50/100      6.45G      1.021     0.7208      1.076         64        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.845      0.925      0.969
0.584

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     51/100       6.5G     0.9581     0.6741      1.014         84        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.896      0.883      0.967
0.584

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100      6.46G     0.9611     0.6753      1.035         59        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.892      0.896      0.977
0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100      6.39G     0.9603     0.6325      1.033         71        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.909      0.935      0.979
0.593

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100      6.45G     0.9564     0.6221       1.04         50        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.4it/s 0.1s
                   all         10         54      0.929      0.959      0.978
0.588

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100      6.44G     0.9311     0.6203      1.027         63        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.0it/s 0.1s
                   all         10         54       0.91      0.949      0.976
0.61

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100      6.48G     0.9512     0.6708      1.043         53        640: 100%
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.866      0.951      0.979
0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     57/100      6.34G     0.9605      0.643      1.055         54        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.863      0.959      0.981
0.6

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     58/100      6.43G     0.9412     0.6408      1.042         65        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.866      0.922      0.974
0.584

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     59/100      6.43G     0.9071     0.6246      1.028         74        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.5it/s 2.8s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.933      0.891      0.974
0.581

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     60/100      6.45G     0.8747     0.5717     0.9989         57        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.7it/s 0.1s
                   all         10         54      0.887      0.846      0.946
0.576

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     61/100       6.4G     0.9082      0.616      1.009         68        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.851      0.908      0.951
0.572

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     62/100      6.43G     0.8993      0.586      0.999         82        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.874      0.945      0.974
0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     63/100      6.48G      0.933     0.6099      1.026         57        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.8it/s 0.1s
                   all         10         54        0.9      0.879      0.979
0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     64/100      6.42G     0.9062     0.6243      1.038         40        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.841      0.919      0.964
0.558

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     65/100      6.35G     0.8942     0.6019      1.011         58        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.9it/s 0.1s
                   all         10         54      0.889       0.87      0.973
0.572

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     66/100      6.41G     0.8377     0.5696     0.9796         57        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.876      0.951      0.982
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     67/100      6.47G      0.856     0.5565     0.9851         60        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54       0.92      0.921      0.979
0.608

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     68/100      6.51G     0.8354     0.5618     0.9859         67        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.916      0.949      0.982
0.603

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     69/100       6.3G     0.8377     0.5441     0.9882         56        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.3it/s 0.1s
                   all         10         54      0.903      0.916      0.976
0.61

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     70/100      6.49G     0.8621     0.5826     0.9882         54        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.911      0.912      0.975
0.588

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     71/100      6.43G     0.8089     0.5185     0.9716         67        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.0it/s 0.1s
                   all         10         54      0.907      0.929      0.964
0.593

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     72/100      6.49G     0.8171     0.5052     0.9571         70        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54      0.911      0.924      0.951
0.563

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     73/100      6.34G     0.8169     0.5463     0.9898         67        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.951      0.903      0.951
0.572

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     74/100      6.41G      0.811     0.5442     0.9764         47        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.947       0.92      0.955
0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     75/100      6.42G     0.7976     0.5309     0.9771         93        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.8it/s 0.1s
                   all         10         54      0.951      0.928      0.972
0.591

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     76/100      6.47G      0.783     0.5348     0.9628         70        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.906      0.939      0.971
0.58

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     77/100      6.34G      0.799     0.5302     0.9841         59        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.3it/s 0.1s
                   all         10         54      0.902      0.932      0.969
0.57

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     78/100      6.42G      0.757     0.5127     0.9494         94        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.4it/s 0.1s
                   all         10         54      0.911      0.915      0.967
0.579

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     79/100      6.47G     0.7327     0.5042     0.9383         59        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.911      0.905      0.964
0.572

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     80/100      6.49G     0.6936      0.451     0.9312         67        640: 100%
```

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━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54       0.915      0.916      0.961
0.574

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     81/100      6.33G     0.7574     0.5288     0.9619         66        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54       0.911      0.913      0.961
0.575

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     82/100       6.5G     0.7412     0.4952     0.9505         66        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54       0.914      0.916      0.968
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     83/100      6.39G     0.7301     0.4963     0.9446         58        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.8it/s 0.1s
                   all         10         54       0.918      0.903      0.969
0.595

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     84/100      6.43G     0.7179     0.4793     0.9416         60        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.7it/s 0.1s
                   all         10         54       0.896      0.916      0.958
0.576

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     85/100      6.34G     0.6973     0.4672     0.9348         56        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.5it/s 0.1s
                   all         10         54       0.888      0.916      0.959
0.576

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     86/100      6.42G     0.7223     0.5006     0.9505         59        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances       Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 7.2it/s 0.1s
                   all         10         54       0.921      0.926      0.976
0.59

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     87/100      6.49G      0.712     0.4795     0.9334         59        640: 100%
```

```
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.6it/s 0.1s
                   all         10         54      0.905      0.906      0.967
0.602

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     88/100       6.5G     0.7176     0.5083     0.9556         51        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.7it/s 0.1s
                   all         10         54      0.921      0.906      0.963
0.59

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     89/100      6.39G     0.6842     0.4625     0.9262         62        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.919      0.907      0.963
0.604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     90/100      6.41G     0.6772     0.4552     0.9255         66        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.2it/s 0.1s
                   all         10         54      0.912       0.91      0.963
0.594
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     91/100       6.4G     0.6548     0.4305     0.9434         36        640: 100%
━━━━━━━━━━━━ 7/7 1.8it/s 3.8s0.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.937      0.922      0.979
0.611

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     92/100       6.4G     0.5899      0.376     0.8955         36        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.931      0.942      0.978
0.6

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     93/100      6.31G     0.5953     0.3694     0.8997         47        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.4it/s 0.1s
                   all         10         54      0.915      0.929      0.961
0.599

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
     94/100      6.4G     0.5976     0.3797     0.9208         38        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.9it/s 0.1s
                   all         10         54      0.916       0.92       0.96
0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     95/100     6.39G     0.5513     0.3534     0.8695         36        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.0it/s 0.1s
                   all         10         54      0.929       0.94      0.974
0.591

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     96/100      6.4G     0.5742     0.3918     0.8919         44        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 8.5it/s 0.1s
                   all         10         54      0.908      0.916       0.96
0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     97/100     6.31G     0.5587     0.3736     0.8941         37        640: 100%
━━━━━━━━━━━━ 7/7 2.7it/s 2.6s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.908      0.924      0.965
0.598

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     98/100      6.4G     0.5799     0.3818     0.8966         36        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 9.6it/s 0.1s
                   all         10         54      0.909      0.919      0.962
0.601

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     99/100     6.41G     0.5608     0.3703     0.8921         39        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.1it/s 0.1s
                   all         10         54      0.908      0.914       0.96
0.6

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
    100/100     6.39G     0.5409      0.373     0.8983         40        640: 100%
━━━━━━━━━━━━ 7/7 2.6it/s 2.7s0.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 10.0it/s 0.1s
                   all         10         54      0.906      0.912       0.96
0.6

100 epochs completed in 0.118 hours.
```

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st.pt, 52.0MB
Optimizer stripped from D:\workplace_00\runs\detect\runs\mnist\v3_no_flip\weights\be
st.pt, 52.0MB

Validating D:\workplace_00\runs\detect\runs\mnist\v3_no_flip\weights\best.pt...
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 5.1it/s 0.2s
                   all         10         54      0.937      0.922      0.979
0.608
                     0          4          6      0.981          1      0.995
0.711
                     1          3          3      0.957          1      0.995
0.588
                     2          7         10       0.99        0.9      0.978
0.543
                     3          6          8          1      0.966      0.995
0.57
                     4          4          4      0.687      0.561      0.895
0.532
                     5          1          1      0.865          1      0.995
0.697
                     6          3          5      0.974          1      0.995
0.587
                     7          3          3      0.928          1      0.995
0.625
                     8          5          7      0.991          1      0.995
0.552
                     9          6          7          1      0.796      0.953
0.676
Speed: 1.3ms preprocess, 15.1ms inference, 0.0ms loss, 0.6ms postprocess per image
Results saved to D:\workplace_00\runs\detect\runs\mnist\v3_no_flip
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 222.244.2 MB/s, size: 11.4 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v3\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all         10         54      0.917      0.903      0.963
0.606
                     0          4          6      0.981          1      0.995
0.711
                     1          3          3      0.956          1      0.995
0.588
                     2          7         10      0.989        0.9      0.978
0.543
                     3          6          8          1      0.968      0.995
0.56
                     4          4          4      0.687      0.562      0.895
0.532
                     5          1          1      0.864          1      0.995
```

```
0.697
                     6          3          5      0.779        0.8       0.83
0.571
                     7          3          3      0.927          1      0.995
0.625
                     8          5          7       0.99          1      0.995
0.563
                     9          6          7          1        0.8      0.953
0.676
Speed: 2.0ms preprocess, 46.7ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val10

===== V3 No-Flip Validation Metrics =====
Precision   : 0.9174
Recall      : 0.9030
mAP50       : 0.9626
mAP50-95    : 0.6064
```

Result Interpretation

The Version 3 dataset achieved the highest precision and the highest mAP50-95 among all experiments. Although its mAP50 was slightly lower than that of Version 2, Version 3 showed the most balanced overall performance and was therefore selected as the final model in this study.

13. Quantitative Comparison of Experimental Results

The table below summarizes the performance of all major experiments. The comparison is structured to highlight the effect of both training-policy modification and dataset-version refinement.

The progression can be interpreted in two stages:

- **Stage 1: Training-policy improvement**
  - V1 baseline → V1 no-flip
- **Stage 2: Dataset-version improvement**
  - V1 no-flip → V2 no-flip → V3 no-flip

This separation is important because it helps explain whether the observed gains came from better augmentation control, better dataset design, or both.

In [2]:

```
from ultralytics import YOLO
import pandas as pd

experiments = {
    "V1 baseline": {
        "weight": "./runs/detect/runs/mnist/v1_baseline/weights/best.pt",
        "data": "./data/mnist/mnist_v1/data_local.yaml",
```

```python
    },
    "V1 no-flip": {
        "weight": "./runs/detect/runs/mnist/v1_no_flip/weights/best.pt",
        "data": "./data/mnist/mnist_v1/data_local.yaml",
    },
    "V2 no-flip": {
        "weight": "./runs/detect/runs/mnist/v2_no_flip/weights/best.pt",
        "data": "./data/mnist/mnist_v2/data_local.yaml",
    },
    "V3 no-flip": {
        "weight": "./runs/detect/runs/mnist/v3_no_flip/weights/best.pt",
        "data": "./data/mnist/mnist_v3/data_local.yaml",
    },
}

rows = []

for exp_name, cfg in experiments.items():
    model = YOLO(cfg["weight"])
    metrics = model.val(
        data=cfg["data"],
        split="val",
        device=0,
        verbose=False
    )

    rows.append({
        "Experiment": exp_name,
        "Precision": round(metrics.box.mp, 4),
        "Recall": round(metrics.box.mr, 4),
        "mAP50": round(metrics.box.map50, 4),
        "mAP50-95": round(metrics.box.map, 4),
    })

results_df = pd.DataFrame(rows)
results_df
```

```
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 102.736.3 MB/s, size: 5.7 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.6s/it 1.6s
                   all         10         54      0.774      0.936      0.964
0.598
Speed: 1.5ms preprocess, 35.6ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val13
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 0.00.0 ms, read: 91.629.8 MB/s, size: 5.5 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v1\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.6s/it 1.6s
                   all         10         54      0.872      0.847      0.966
0.6
Speed: 0.8ms preprocess, 32.9ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val14
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 7.215.7 ms, read: 2.11.3 MB/s, size: 9.9 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v2\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all         10         54       0.85      0.898       0.97
0.588
Speed: 0.7ms preprocess, 26.3ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val15
Ultralytics 8.4.22  Python-3.11.9 torch-2.10.0+cu126 CUDA:0 (NVIDIA GeForce RTX 406
0, 8188MiB)
Model summary (fused): 93 layers, 25,845,550 parameters, 0 gradients, 78.7 GFLOPs
val: Fast image access  (ping: 5.211.3 ms, read: 4.22.7 MB/s, size: 10.2 KB)
val: Scanning D:\workplace_00\data\mnist\mnist_v3\valid\labels.cache... 10 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10/10  0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all         10         54      0.917      0.903      0.963
0.606
Speed: 0.7ms preprocess, 27.1ms inference, 0.0ms loss, 0.4ms postprocess per image
Results saved to D:\workplace_00\runs\detect\val16
```

Out[2]:

	Experiment	Precision	Recall	mAP50	mAP50-95
0	V1 baseline	0.7736	0.9357	0.9636	0.5978
1	V1 no-flip	0.8715	0.8472	0.9658	0.6004
2	V2 no-flip	0.8504	0.8976	0.9695	0.5882
3	V3 no-flip	0.9174	0.9030	0.9626	0.6064

The final comparison shows that each experiment improved a different aspect of object detection performance. Removing flip augmentation increased precision and slightly improved the overall mAP scores, indicating that task-specific augmentation control was beneficial. Version 2 achieved the highest mAP50 and strong recall, suggesting better general detection coverage. Version 3 achieved the highest precision and mAP50-95, indicating the best overall balance between reliability and localization quality.

The improvement from V2 to V3 suggests that adding appearance-based augmentations such as brightness, exposure, and blur contributed to a more reliable final detector, especially in terms of precision and localization-aware performance.

14. Qualitative Analysis of Prediction Results

In addition to numerical evaluation, prediction results on the test images were saved for qualitative analysis.

This step is important because object detection performance cannot be fully understood through numerical metrics alone. By visually inspecting prediction outputs, it becomes possible to identify:

- successful detections
- missed detections
- false positives
- duplicate boxes
- class confusion between visually similar digits

The prediction outputs of each experiment are displayed below for visual comparison.

In [2]:

```
from ultralytics import YOLO

models = {
    "v1_baseline": "./runs/detect/runs/mnist/v1_baseline/weights/best.pt",
    "v1_no_flip": "./runs/detect/runs/mnist/v1_no_flip/weights/best.pt",
    "v2_no_flip": "./runs/detect/runs/mnist/v2_no_flip/weights/best.pt",
    "v3_no_flip": "./runs/detect/runs/mnist/v3_no_flip/weights/best.pt",
}

test_sources = {
```

```
    "v1_baseline": "./data/mnist/mnist_v1/test/images",
    "v1_no_flip": "./data/mnist/mnist_v1/test/images",
    "v2_no_flip": "./data/mnist/mnist_v2/test/images",
    "v3_no_flip": "./data/mnist/mnist_v3/test/images",
}

for exp_name, model_path in models.items():
    print(f"\n===== Predicting: {exp_name} =====")
    model = YOLO(model_path)
    model.predict(
        source=test_sources[exp_name],
        conf=0.25,
        save=True,
        device=0,
        project="./runs/mnist_preds",
        name=exp_name,
        exist_ok=True
    )

print("\n모든 예측 이미지 저장 완료")
```

```
===== Predicting: v1_baseline =====

image 1/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000006_jpg.rf.517a38906e1d
34cb6aae423489bc1b4a.jpg: 640x640 2 0s, 1 6, 12.9ms
image 2/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000022_jpg.rf.d89e1085766e
248c0322d57d5f1afde2.jpg: 640x640 2 1s, 1 2, 1 3, 2 4s, 1 5, 1 8, 1 9, 13.5ms
image 3/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000028_jpg.rf.52390ccbb7f3
433856c47868f3123877.jpg: 640x640 1 0, 1 2, 1 4, 12.2ms
image 4/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000029_jpg.rf.13d22c3ff913
0850dd6e7f5b732d195f.jpg: 640x640 1 1, 4 2s, 1 3, 1 5, 4 6s, 1 8, 12.7ms
image 5/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000031_jpg.rf.fa82149c089a
2e46241af4e4d1e8b013.jpg: 640x640 1 0, 1 2, 1 6, 1 7, 1 8, 12.6ms
Speed: 2.0ms preprocess, 12.8ms inference, 1.0ms postprocess per image at shape (1,
3, 640, 640)
Results saved to D:\workplace_00\runs\detect\runs\mnist_preds\v1_baseline

===== Predicting: v1_no_flip =====

image 1/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000006_jpg.rf.517a38906e1d
34cb6aae423489bc1b4a.jpg: 640x640 1 0, 1 6, 12.6ms
image 2/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000022_jpg.rf.d89e1085766e
248c0322d57d5f1afde2.jpg: 640x640 2 1s, 1 2, 1 3, 1 4, 1 9, 13.3ms
image 3/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000028_jpg.rf.52390ccbb7f3
433856c47868f3123877.jpg: 640x640 1 0, 1 1, 1 2, 1 4, 12.9ms
image 4/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000029_jpg.rf.13d22c3ff913
0850dd6e7f5b732d195f.jpg: 640x640 1 1, 3 2s, 1 4, 2 6s, 1 8, 12.7ms
image 5/5 d:\workplace_00\data\mnist\mnist_v1\test\images\000031_jpg.rf.fa82149c089a
2e46241af4e4d1e8b013.jpg: 640x640 1 0, 1 2, 1 6, 1 7, 1 8, 12.4ms
Speed: 2.1ms preprocess, 12.8ms inference, 0.9ms postprocess per image at shape (1,
3, 640, 640)
Results saved to D:\workplace_00\runs\detect\runs\mnist_preds\v1_no_flip

===== Predicting: v2_no_flip =====

image 1/5 d:\workplace_00\data\mnist\mnist_v2\test\images\000006_jpg.rf.ff1519060ffa
c9999a9b5f57fb4de5c7.jpg: 640x640 1 0, 1 6, 1 8, 12.7ms
image 2/5 d:\workplace_00\data\mnist\mnist_v2\test\images\000022_jpg.rf.057b1a75f711
97b37f9b21fa28b9c45b.jpg: 640x640 2 1s, 1 2, 1 3, 1 4, 1 9, 12.8ms
image 3/5 d:\workplace_00\data\mnist\mnist_v2\test\images\000028_jpg.rf.996b6389031c
739ace5c2560d7adbcd6.jpg: 640x640 1 0, 1 2, 1 4, 12.4ms
image 4/5 d:\workplace_00\data\mnist\mnist_v2\test\images\000029_jpg.rf.846a2c69bc88
15646c66e5b14858eca6.jpg: 640x640 1 1, 5 2s, 1 3, 3 6s, 1 8, 12.6ms
image 5/5 d:\workplace_00\data\mnist\mnist_v2\test\images\000031_jpg.rf.04905887d22b
35b6cbdceb79c0365c6f.jpg: 640x640 2 0s, 1 2, 1 7, 1 8, 12.3ms
Speed: 1.5ms preprocess, 12.6ms inference, 0.9ms postprocess per image at shape (1,
3, 640, 640)
Results saved to D:\workplace_00\runs\detect\runs\mnist_preds\v2_no_flip

===== Predicting: v3_no_flip =====

image 1/5 d:\workplace_00\data\mnist\mnist_v3\test\images\000006_jpg.rf.ff1519060ffa
c9999a9b5f57fb4de5c7.jpg: 640x640 1 0, 2 6s, 1 8, 11.8ms
image 2/5 d:\workplace_00\data\mnist\mnist_v3\test\images\000022_jpg.rf.057b1a75f711
97b37f9b21fa28b9c45b.jpg: 640x640 2 1s, 1 2, 1 3, 1 4, 1 9, 12.6ms
image 3/5 d:\workplace_00\data\mnist\mnist_v3\test\images\000028_jpg.rf.996b6389031c
739ace5c2560d7adbcd6.jpg: 640x640 1 0, 1 2, 1 4, 1 9, 11.8ms
```

```
image 4/5 d:\workplace_00\data\mnist\mnist_v3\test\images\000029_jpg.rf.846a2c69bc88
15646c66e5b14858eca6.jpg: 640x640 1 1, 3 2s, 2 3s, 1 4, 2 6s, 1 8, 12.0ms
image 5/5 d:\workplace_00\data\mnist\mnist_v3\test\images\000031_jpg.rf.04905887d22b
35b6cbdceb79c0365c6f.jpg: 640x640 1 0, 1 2, 1 3, 1 6, 1 7, 11.8ms
Speed: 1.6ms preprocess, 12.0ms inference, 0.9ms postprocess per image at shape (1,
3, 640, 640)
Results saved to D:\workplace_00\runs\detect\runs\mnist_preds\v3_no_flip

모든 예측 이미지 저장 완료
```

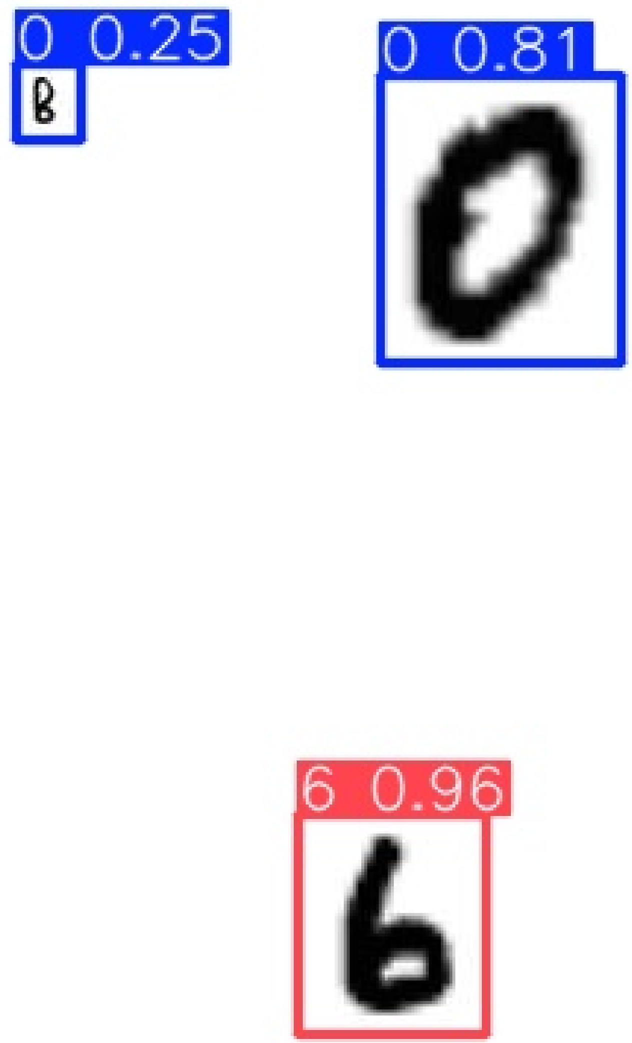
In [3]:

```python
from pathlib import Path
from IPython.display import display, Markdown, Image

pred_dirs = {
    "V1 Baseline": Path("./runs/detect/runs/mnist_preds/v1_baseline"),
    "V1 No-Flip": Path("./runs/detect/runs/mnist_preds/v1_no_flip"),
    "V2 No-Flip": Path("./runs/detect/runs/mnist_preds/v2_no_flip"),
    "V3 No-Flip": Path("./runs/detect/runs/mnist_preds/v3_no_flip"),
}

image_exts = {".jpg", ".jpeg", ".png", ".bmp", ".webp"}

for title, folder in pred_dirs.items():
    display(Markdown(f"## {title}"))
    print("folder:", folder.resolve(), "exists:", folder.exists())

    if not folder.exists():
        print("폴더가 없습니다.\n")
        continue

    image_files = sorted([p for p in folder.iterdir() if p.suffix.lower() in image_

    if not image_files:
        print("표시할 이미지가 없습니다.\n")
        continue

    for img_path in image_files:
        display(Markdown(f"**{img_path.name}**"))
        display(Image(filename=str(img_path), width=700))
```

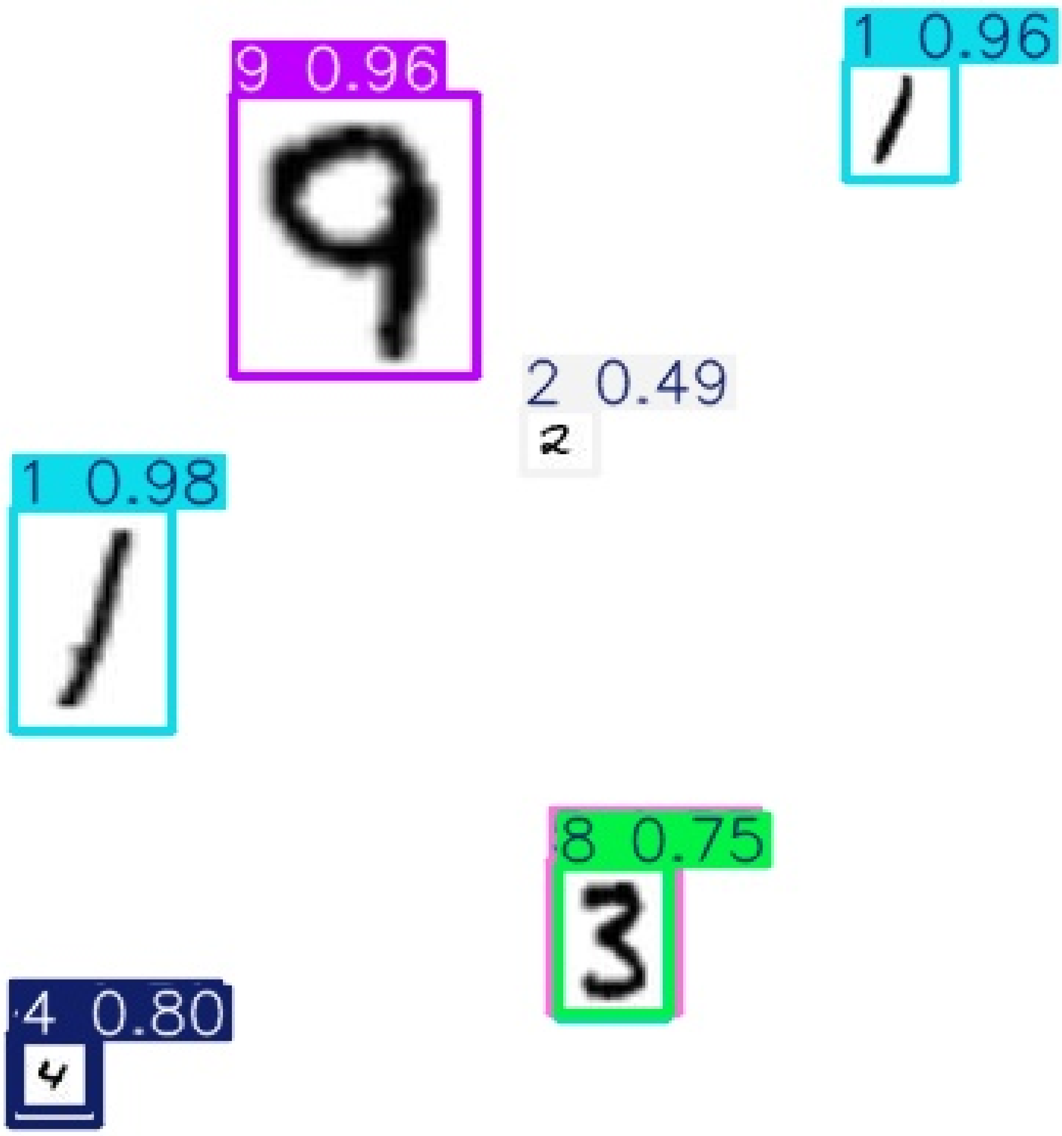
V1 Baseline

```
folder: D:\workplace_00\runs\detect\runs\mnist_preds\v1_baseline exists: True
```

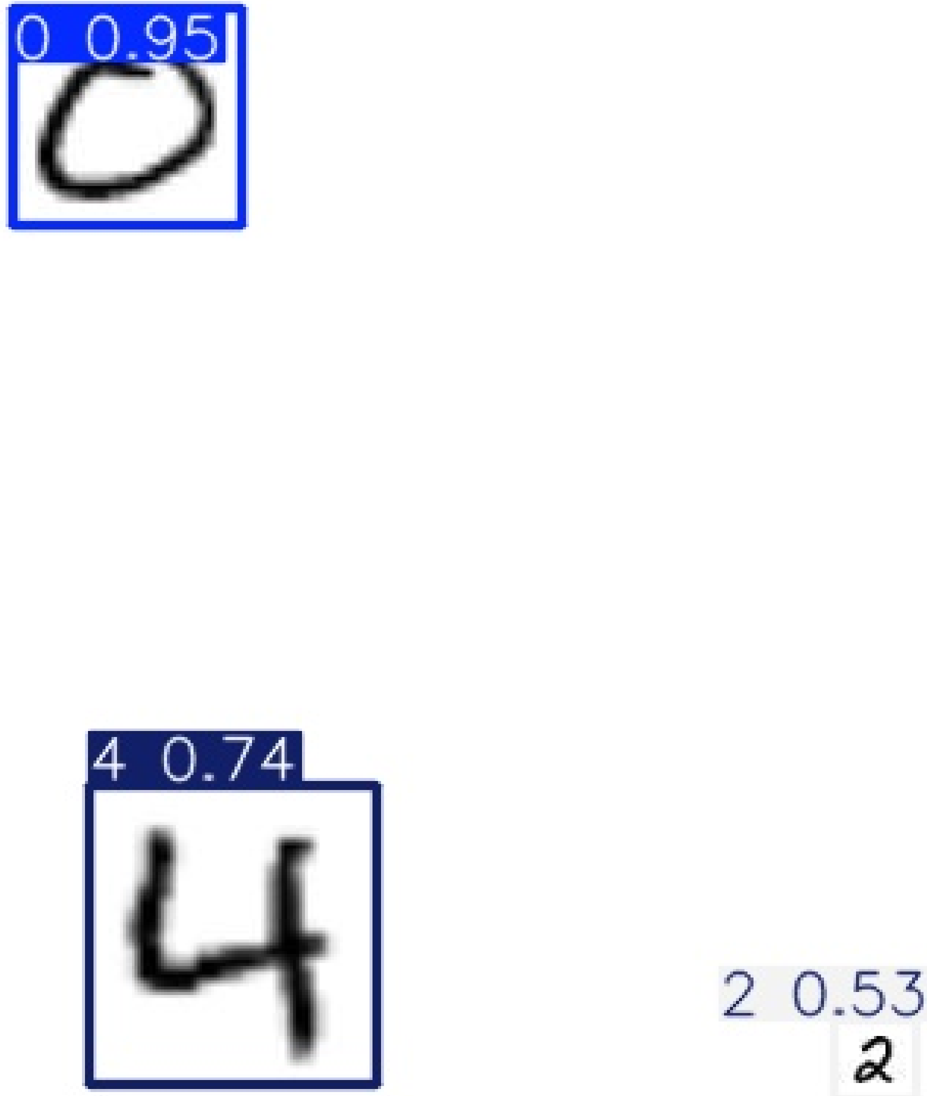
000006_jpg.rf.517a38906e1d34cb6aae423489bc1b4a.jpg



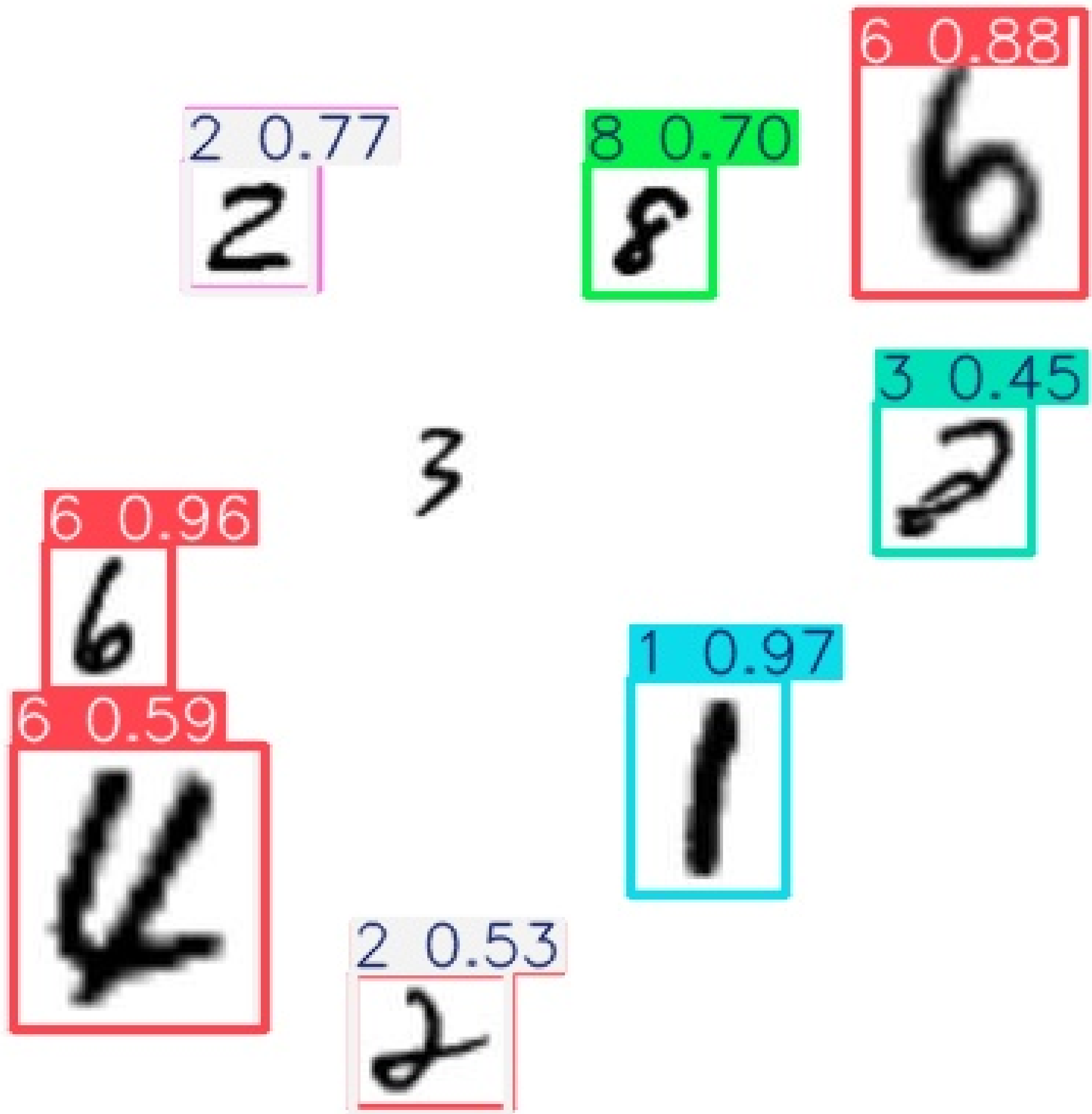
000022_jpg.rf.d89e1085766e248c0322d57d5f1afde2.jpg



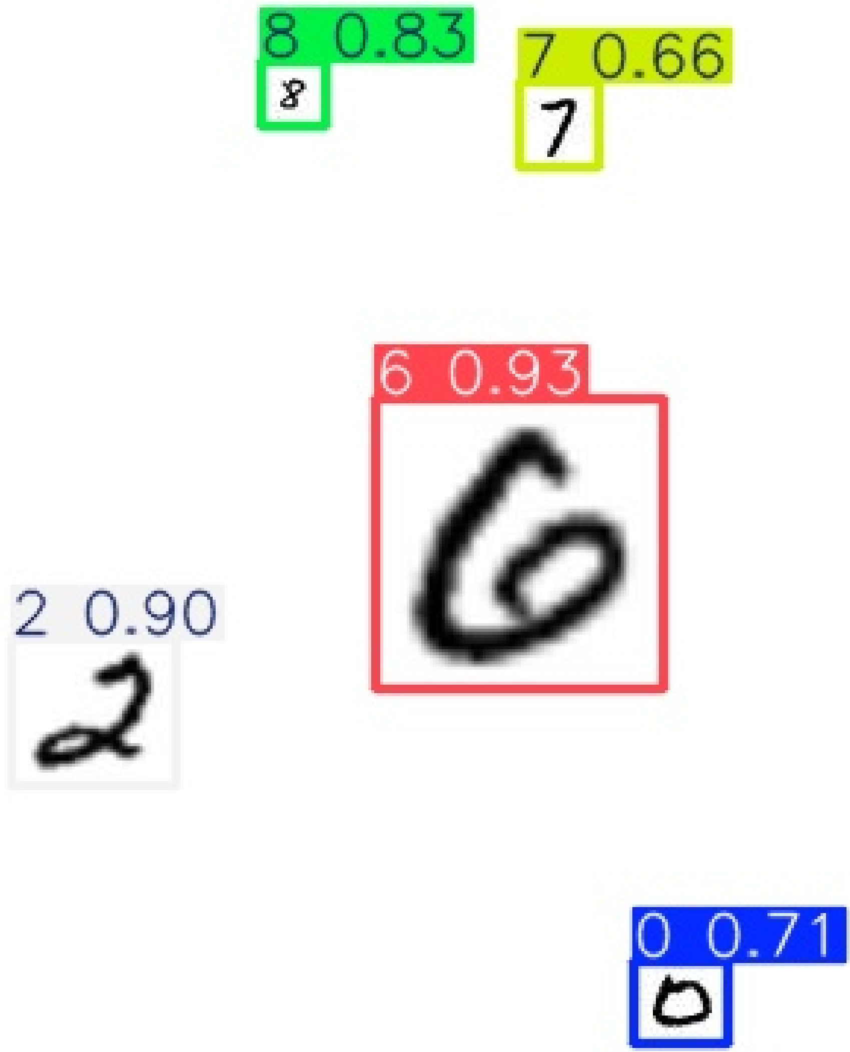
000028_jpg.rf.52390ccbb7f3433856c47868f3123877.jpg



000029_jpg.rf.13d22c3ff9130850dd6e7f5b732d195f.jpg



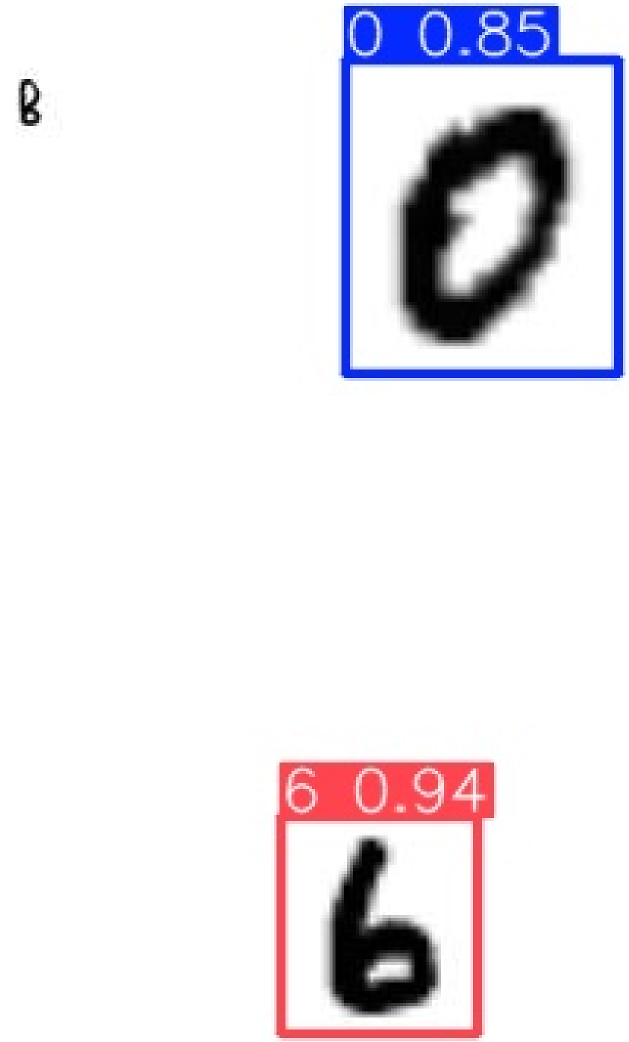
000031_jpg.rf.fa82149c089a2e46241af4e4d1e8b013.jpg



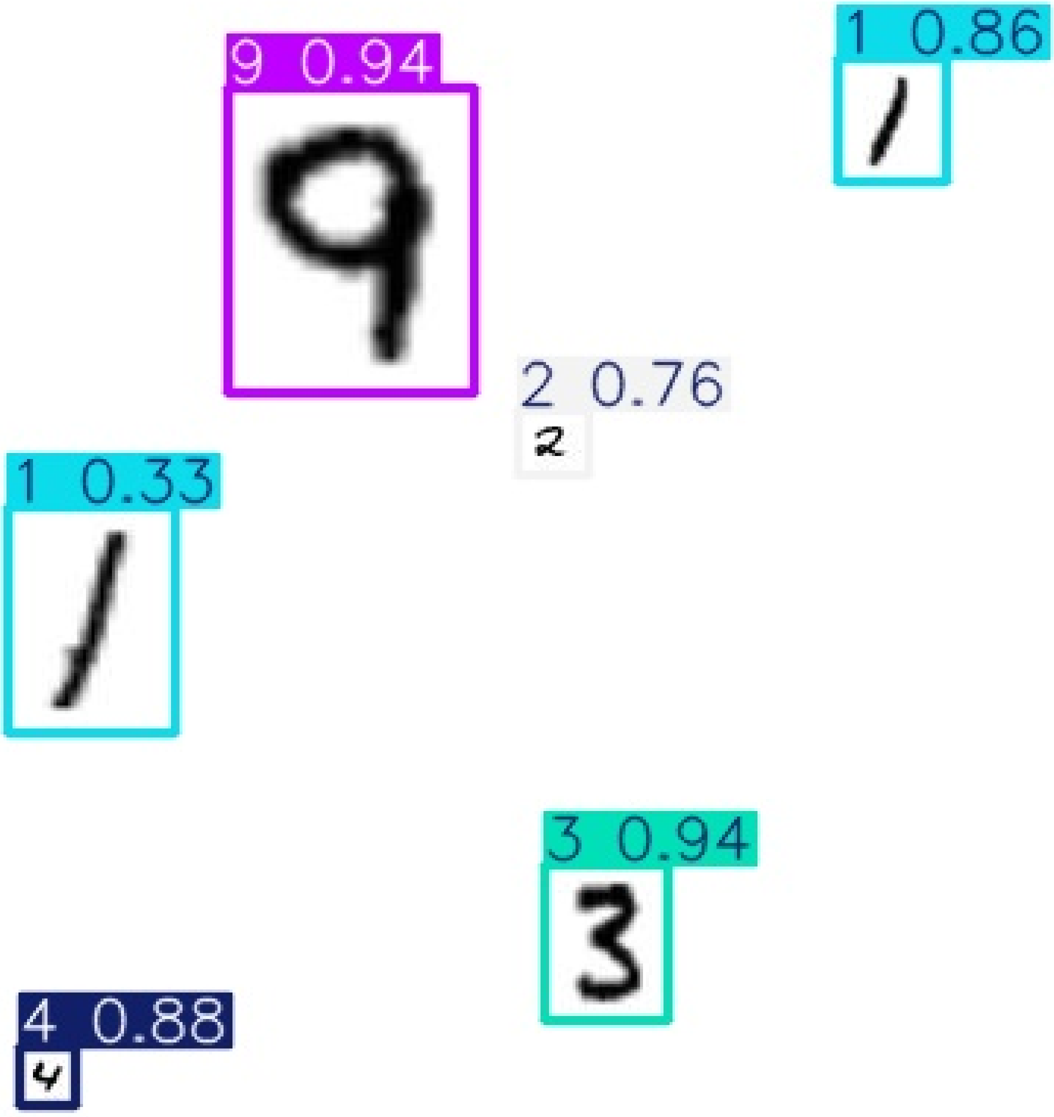
V1 No-Flip

folder: D:\workplace_00\runs\detect\runs\mnist_preds\v1_no_flip exists: True

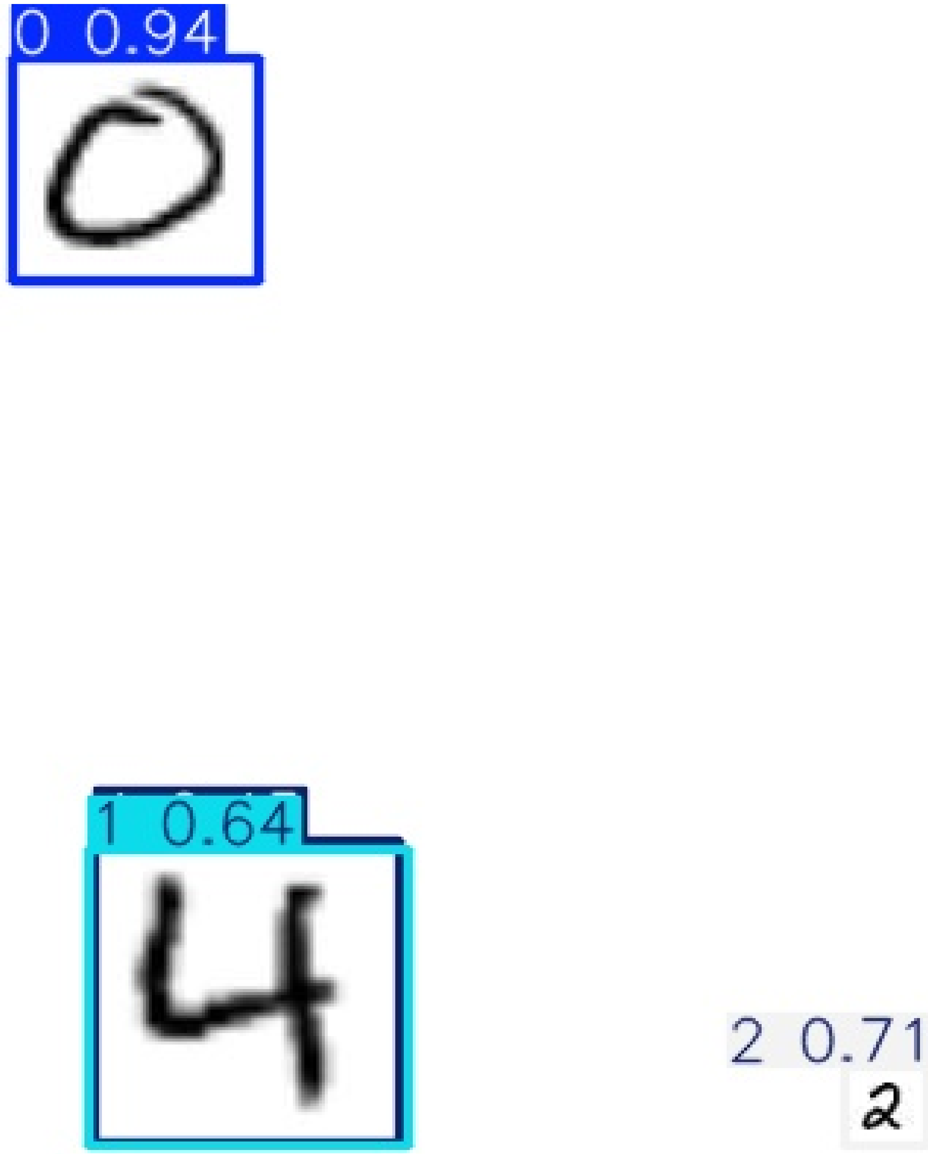
000006_jpg.rf.517a38906e1d34cb6aae423489bc1b4a.jpg



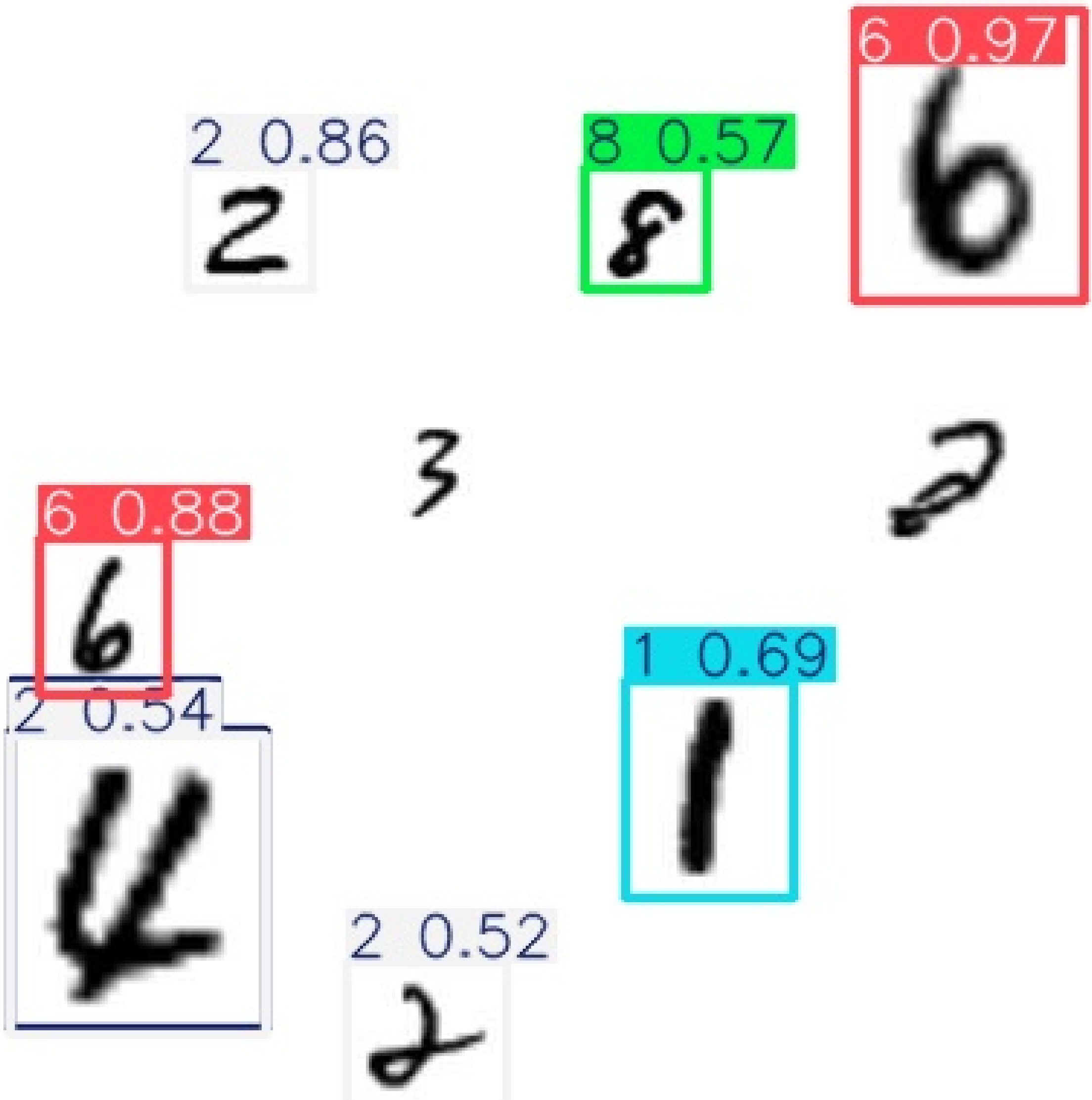
000022_jpg.rf.d89e1085766e248c0322d57d5f1afde2.jpg



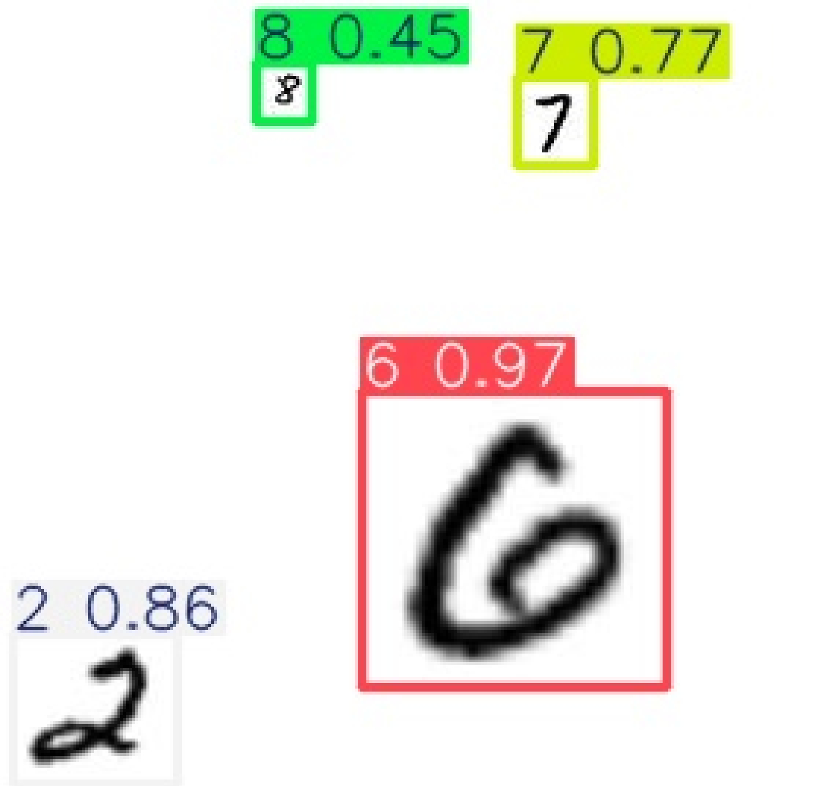
000028_jpg.rf.52390ccbb7f3433856c47868f3123877.jpg



000029_jpg.rf.13d22c3ff9130850dd6e7f5b732d195f.jpg



000031_jpg.rf.fa82149c089a2e46241af4e4d1e8b013.jpg

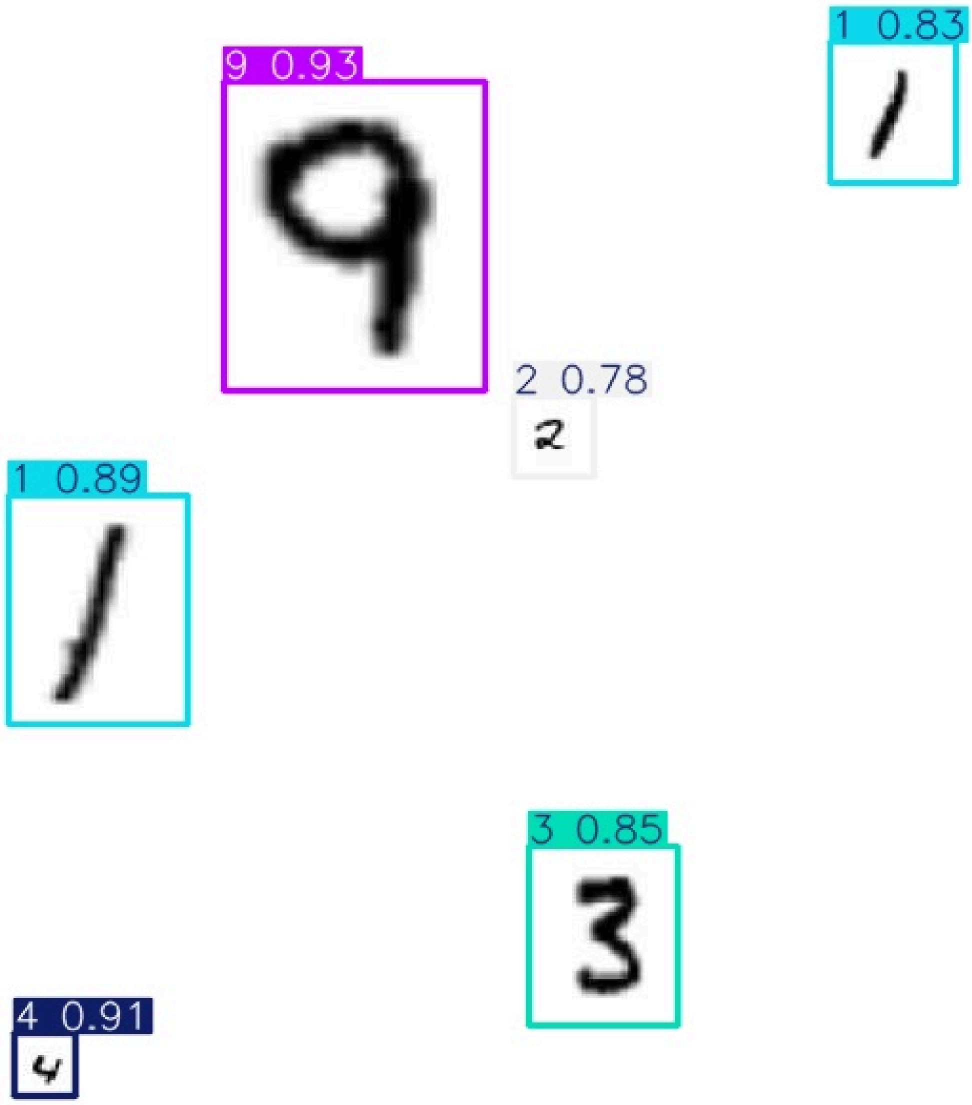


V2 No-Flip

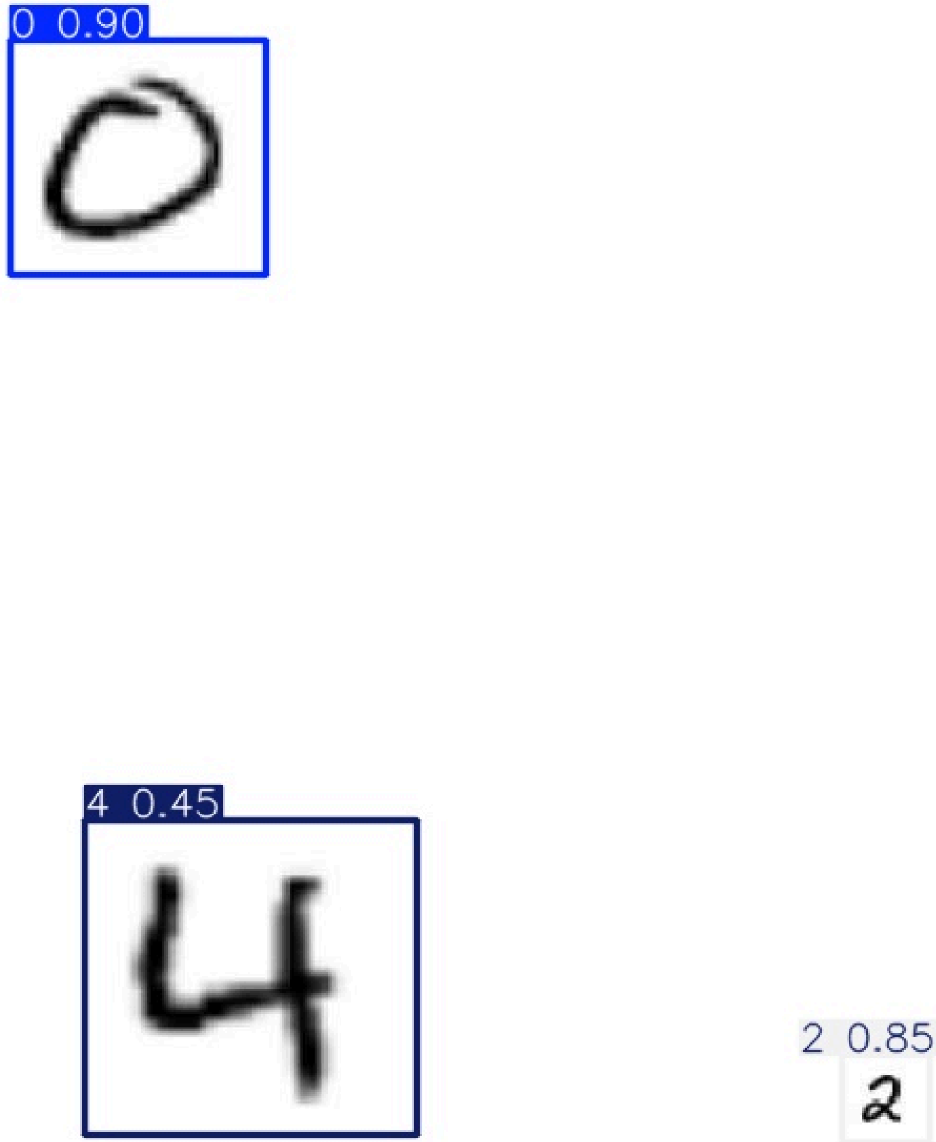
```
folder: D:\workplace_00\runs\detect\runs\mnist_preds\v2_no_flip exists: True
```

000006_jpg.rf.ff1519060ffac9999a9b5f57fb4de5c7.jpg

000022_jpg.rf.057b1a75f71197b37f9b21fa28b9c45b.jpg



000028_jpg.rf.996b6389031c739ace5c2560d7adbcd6.jpg



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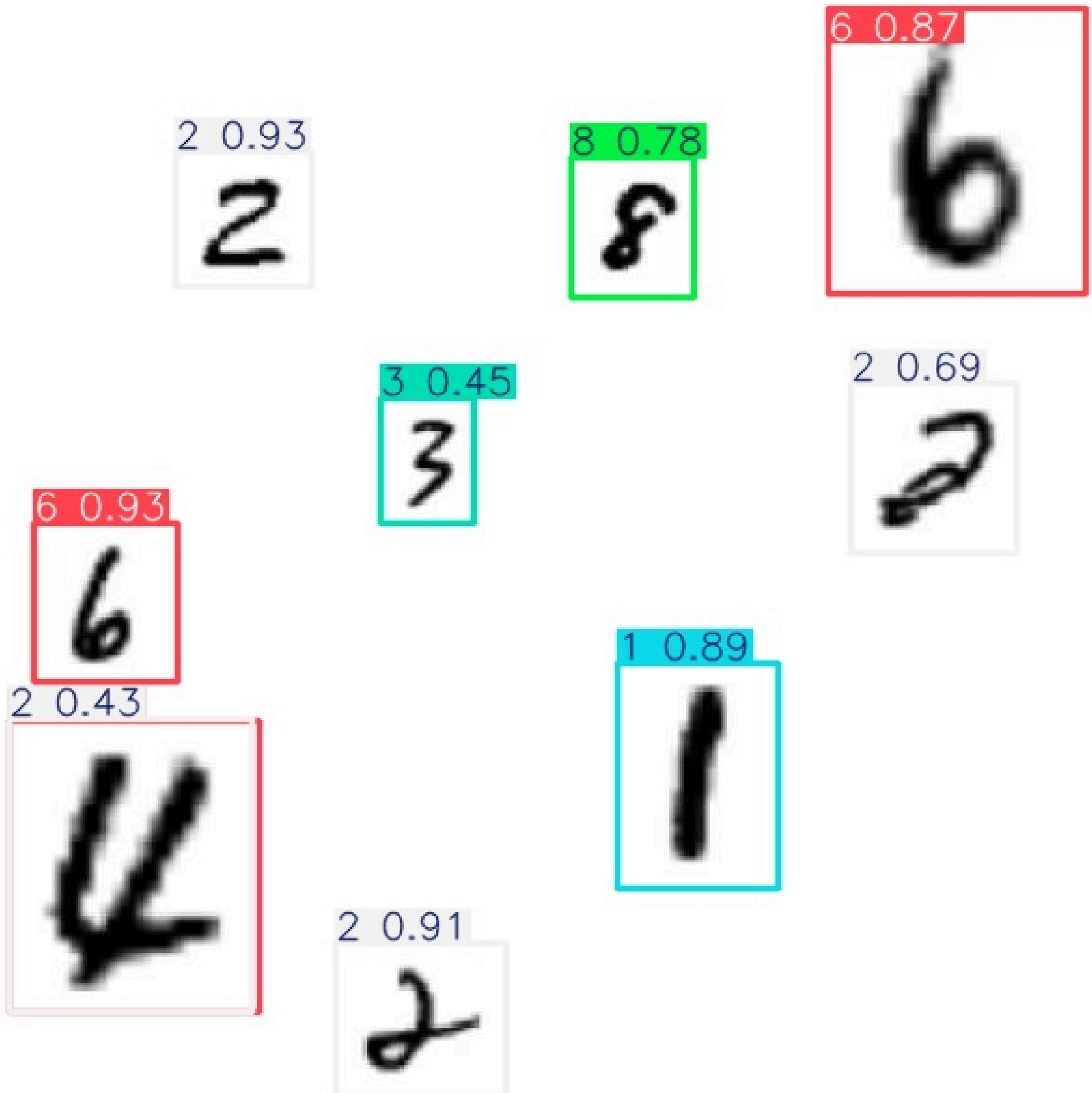

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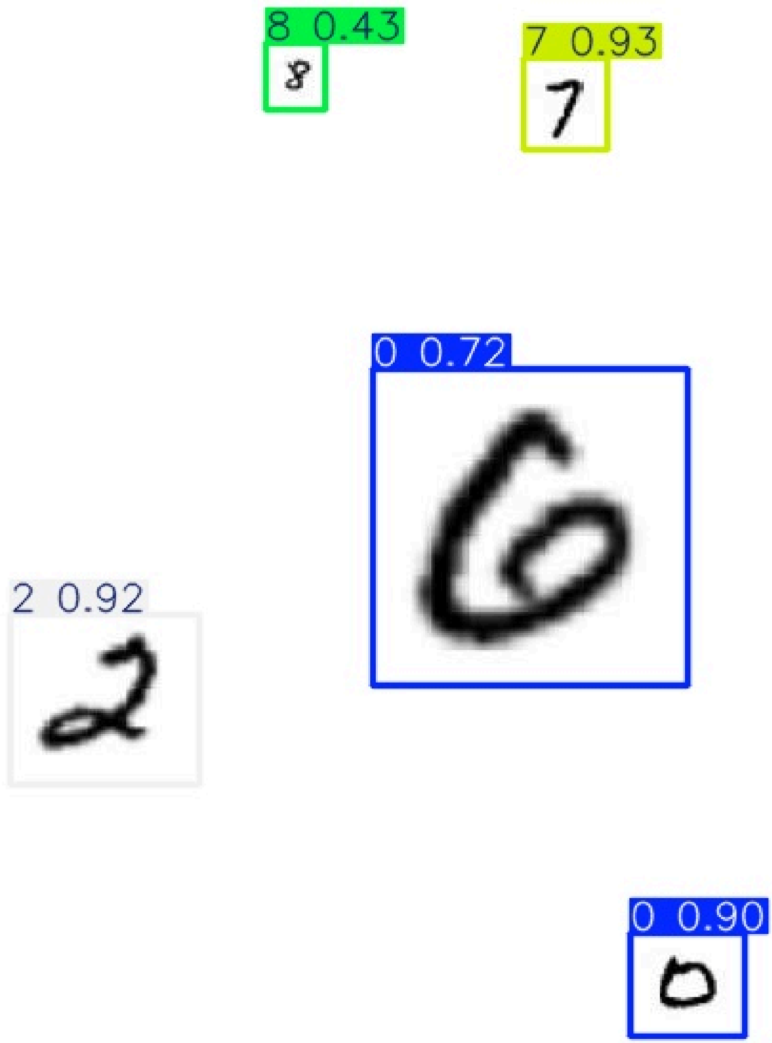

V3 No-Flip

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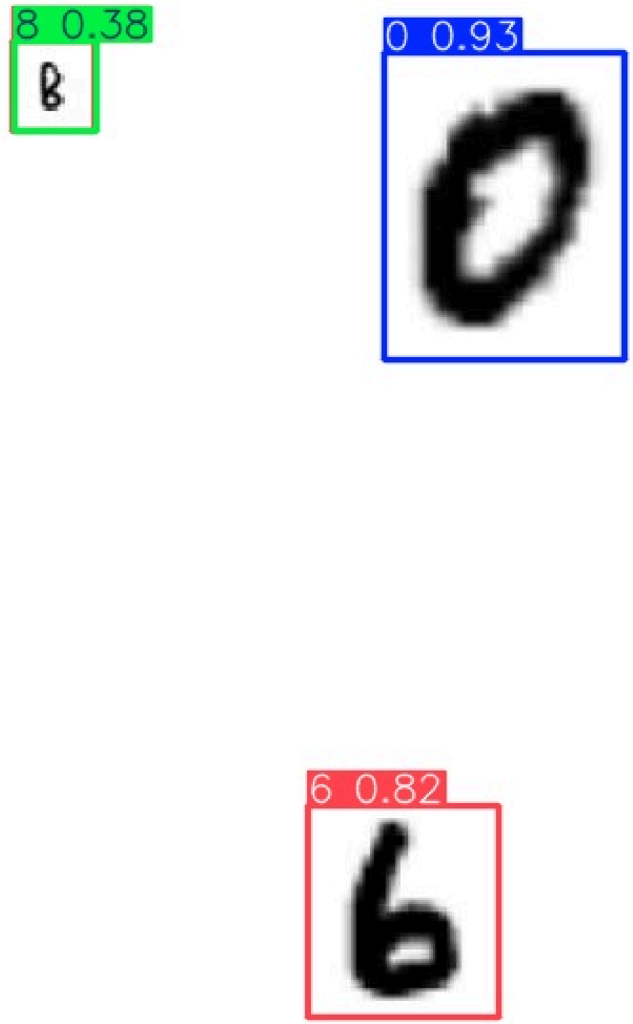

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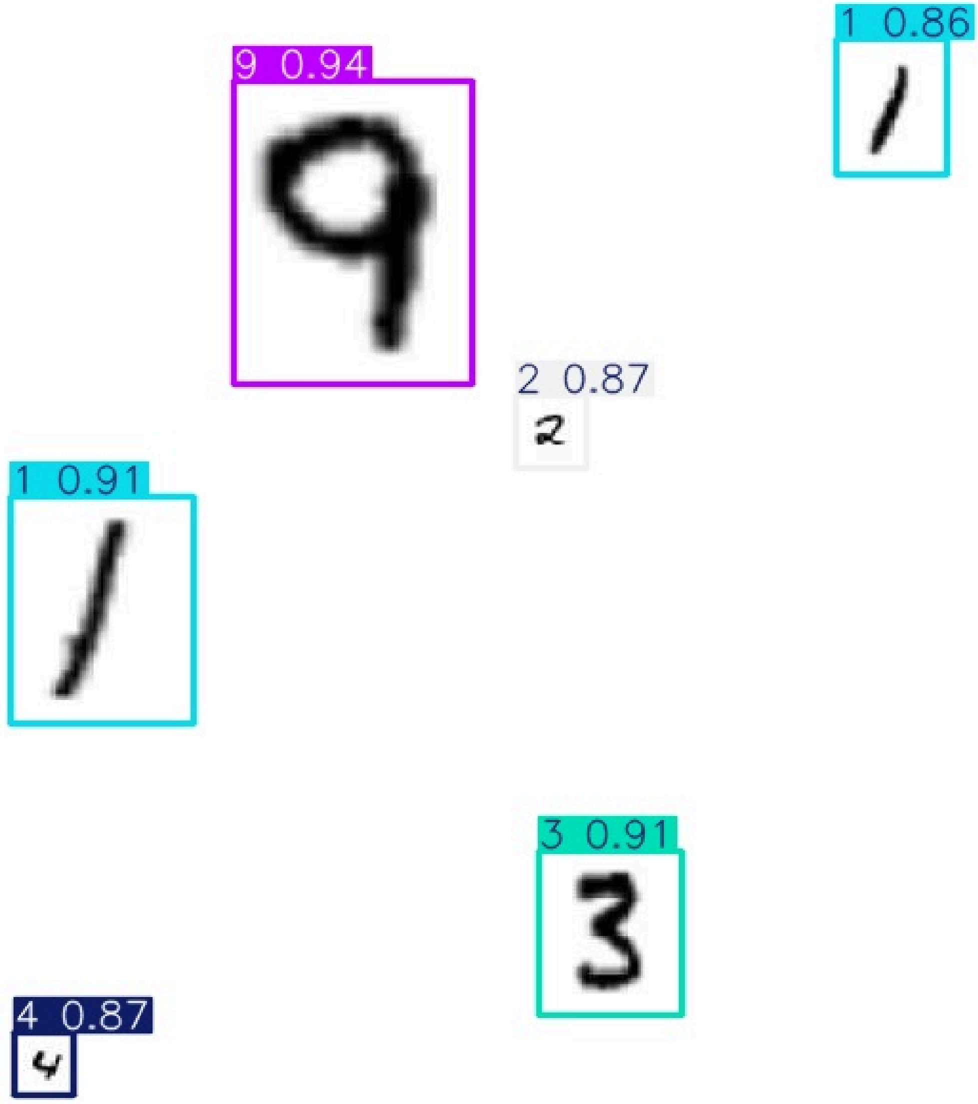

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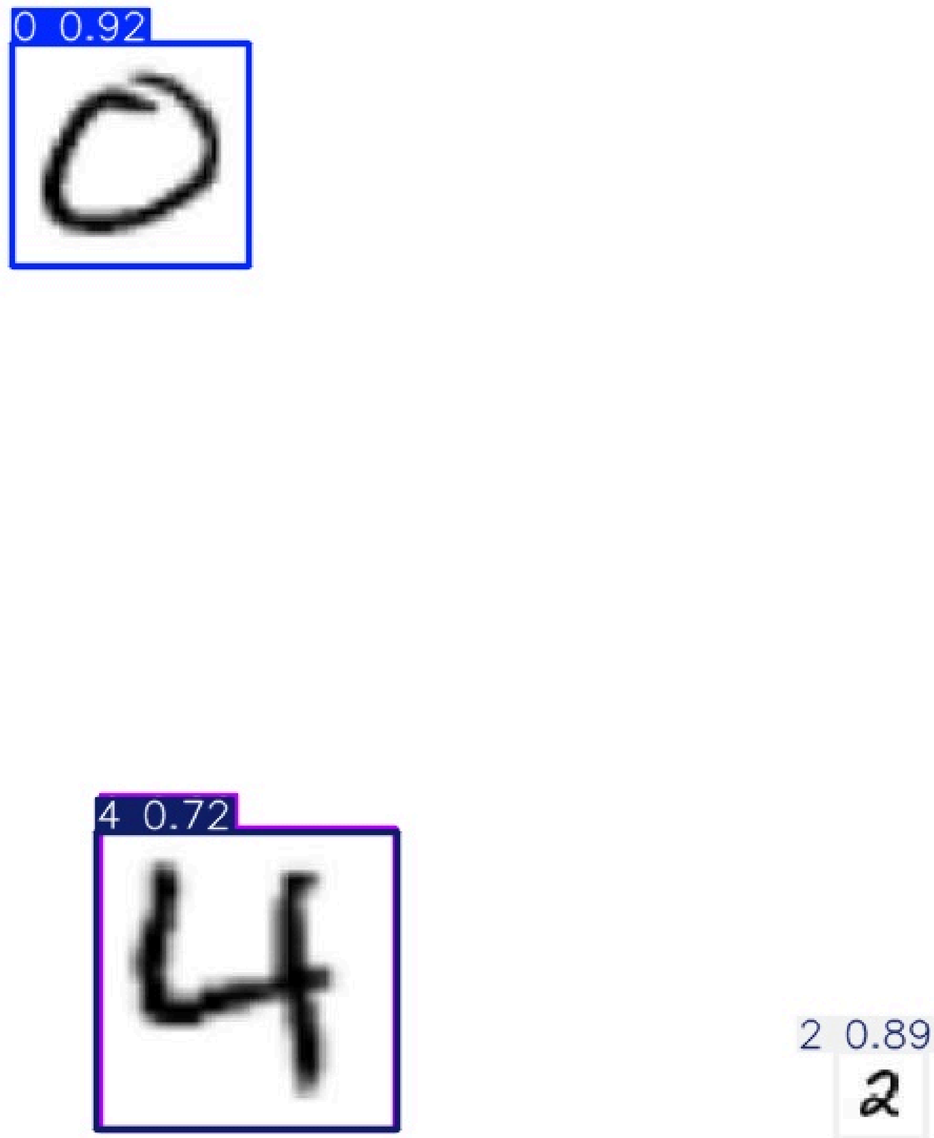

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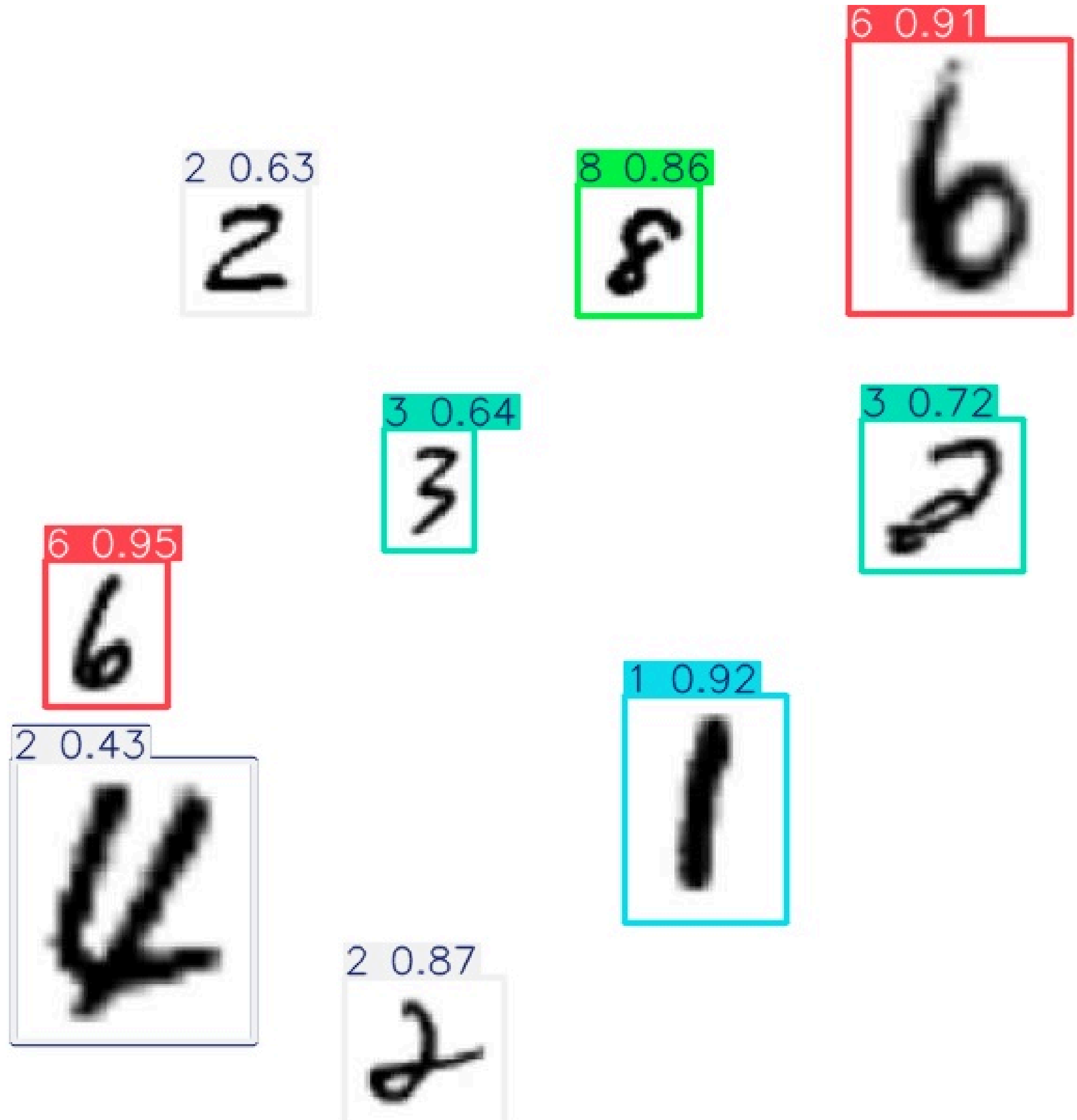

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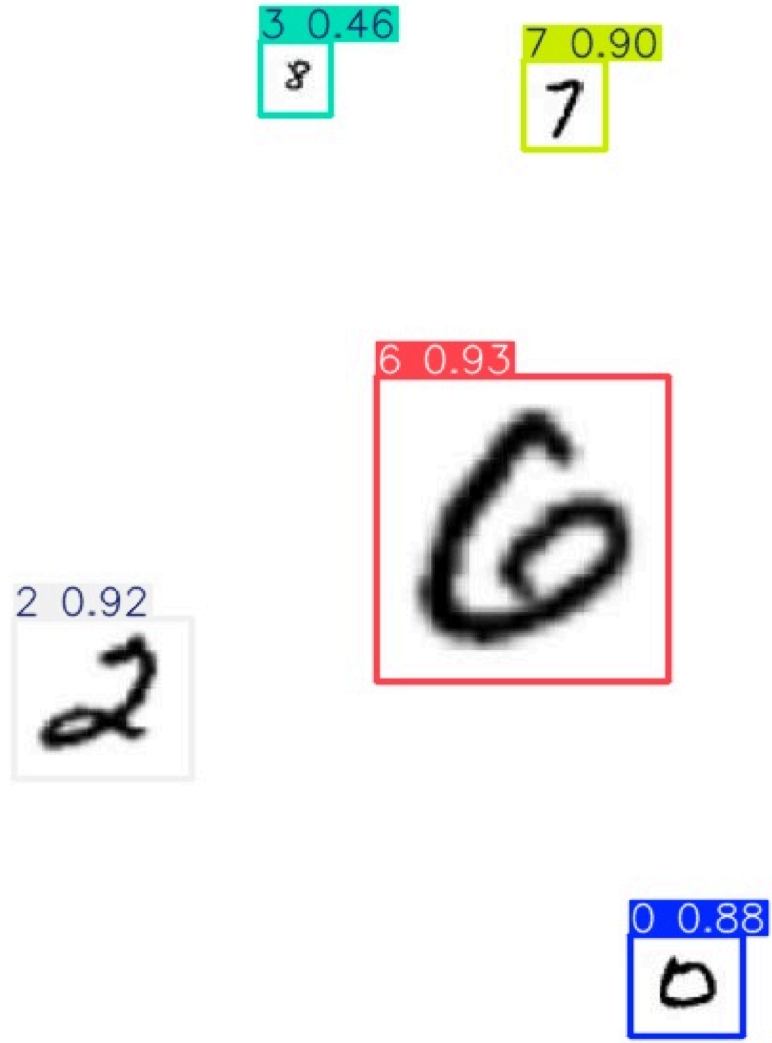

The quantitative and qualitative results were consistent with each other. The numerical improvements in precision and mAP were also reflected in the prediction outputs, where later models showed fewer incorrect detections and more stable object localization. This alignment between metrics and visual results strengthens the reliability of the final experimental conclusion.

Discussion

The experimental results show a clear and meaningful improvement trend across the full development process.

First, the comparison between the baseline model and the no-flip version of V1 confirmed that generic augmentation strategies are not always suitable for handwritten digit detection. Removing flip augmentation increased precision and slightly improved both mAP50 and mAP50-95, although recall decreased. This suggests that the detector became more conservative but also more reliable, which is consistent with the semantic sensitivity of handwritten digits to horizontal flipping.

Second, the Roboflow-based dataset versioning strategy contributed to further model improvement. Version 2 achieved the highest **mAP50 (0.9695)** and strong recall, indicating that the augmented dataset improved the model's general detection capability and object coverage. This means the model was able to detect more digit instances successfully under the standard IoU criterion.

However, Version 3 produced the strongest final overall result. Although its mAP50 was slightly lower than that of Version 2, it achieved the highest **Precision (0.9174)** and the highest **mAP50-95 (0.6064)** among all experiments. These results indicate that Version 3 provided the best balance between detection reliability and localization quality.

The quantitative and qualitative results were broadly consistent with each other. The later experiments showed not only improved numerical performance but also more stable detection behavior in the saved prediction images. This consistency strengthens the conclusion that both dataset refinement and augmentation control played an important role in improving the final detector.

Conclusion

This project successfully demonstrated the development and improvement process of an MNIST-based object detection model using Roboflow and YOLO.

The workflow included:

1. dataset inspection and local configuration
2. baseline training
3. identification of an unsuitable augmentation strategy
4. retraining without flip augmentation
5. training on improved dataset versions
6. quantitative and qualitative comparison

The final results showed that different experiments improved different aspects of performance. The no-flip V1 model demonstrated the importance of task-specific augmentation control. Version 2 improved general detection performance and achieved the highest mAP50. Version 3 achieved the best overall balance and was selected as the final model because it produced the highest precision and the highest mAP50-95.

Overall, this study shows that object detection performance can be improved not only through model training itself, but also through careful dataset versioning, appropriate augmentation design, and systematic experimental comparison.

Limitations and Future Work

This study also has several limitations.

First, the validation and test sets were relatively small, so the evaluation results should be interpreted primarily as comparative rather than absolute. Second, the experiments were conducted on a specialized handwritten digit dataset rather than a natural image detection benchmark.

For future work, the following extensions are recommended:

- increasing the dataset size
- testing larger YOLO models
- performing additional hyperparameter tuning
- analyzing class-wise failure cases in more detail
- evaluating the model on newly generated composite digit scenes